

Suppose next that there is a small error  $y_k \varepsilon_k$  in *each* entry of the table of function values. The result will be the correct difference table plus a sum of difference tables like the one with a single error. Thus the error will be the sum of the errors in the individual tables. To find this expression we apply the above formula to get

$$\text{Error in } \{\Delta^k y_0\} = \sum_{m=0}^k (-1)^m C(k, m) y_{k-m} \varepsilon_{k-m}$$

If the  $|y_{k-m} \varepsilon_{k-m}|$  are bounded by  $\varepsilon$ , then the error is bounded by

$$\left| \sum_{m=0}^k C(k, m) \right| \varepsilon = (1 + 1)^k \varepsilon = 2^k \varepsilon$$

This bound can be attained if all the errors are the same size but with alternating sign, that is, if

$$y_{k-m} \varepsilon_{k-m} (-1)^m = \varepsilon$$

Most of the time the bound will not be attained and is very pessimistic.

#### 10.4 THE STATISTICS OF ROUNDOFF

Since absolute bounds on the roundoff are apt to be very pessimistic, it is often necessary to resort to more realistic, but admittedly more dangerous, statistical estimates.

The roundoff of a single number  $y_i$  is distributed equilikely in the interval  $-1/2 \leq x < 1/2$  of the last digit kept. In a continuing product we will have

$$y_1(1 + \varepsilon_1)y_2(1 + \varepsilon_2) \cdots y_n(1 + \varepsilon_n)$$

and neglecting the product of two or more  $\varepsilon_i$ , we find that this is

$$y_1 y_2 \cdots y_n [1 + (\varepsilon_1 + \varepsilon_2 + \cdots + \varepsilon_n)]$$

By the central-limit theorem a sum of  $n$  independent random numbers approaches the normal distribution as  $n$  approaches infinity,

$$\frac{1}{\sqrt{2\pi}} e^{-x^2}$$

A similar effect applies for divisions except that we use

$$\frac{1}{1 + \varepsilon_i} = 1 - \varepsilon_i$$

and the corresponding  $\varepsilon_i$  appear with minus signs.

For addition and subtraction there is a shift before the mantissas are combined. It is more difficult to say exactly what happens, but the general effect is much the same. While the individual roundoffs are from a flat distribution, the roundoff effect of many arithmetic operations *tends* toward a normal distribution. This is readily verified experimentally.

In a difference table the bound we obtained was again very pessimistic. What is reasonable to expect?

To apply statistics in this case we need first to form a model of the situation we are estimating. The roundoff in a single table is perfectly definite, but we *imagine* that it is one table drawn from an *ensemble* of similar tables but with varying roundoffs. It is natural to suppose that negative and positive roundoffs are equally likely and that the roundoffs at different places in the table are independent. In mathematical notation these assumptions mean that

$$\begin{aligned} \text{Av}\{\varepsilon_i\} &= 0 \\ \text{Av}\{\varepsilon_i \varepsilon_j\} &= 0 \quad i \neq j \end{aligned}$$

The assumption of independence of the roundoff at successive positions is sometimes dangerous, especially in recursive calculations where the computation for one value depends upon that of the previous value or values.

When we apply our statistical averages, we average the  $\varepsilon_i$  over the ensemble of tables, *not* over the index  $i$ , in the following formulas. We have

$$\begin{aligned} \text{Av}\{y_i \varepsilon_i\} &= y_i \text{Av}\{\varepsilon_i\} = 0 \\ \text{Variance}\{y_i \varepsilon_i\} &= \text{Var}\{y_i \varepsilon_i\} \\ &= \text{Av}\{y_i^2 \varepsilon_i^2\} = \sigma^2 \end{aligned}$$

where  $\sigma^2$  is some number which we will later examine more closely and which is *assumed* to be independent of  $i$ .

For the  $k$ th differences in a table we have correspondingly

$$\begin{aligned} \text{Av}\{\Delta^k y_i \varepsilon_i\} &= \text{Av}\left\{ \sum_{m=0}^k (-1)^m C(k, m) y_{k-m} \varepsilon_{k-m} \right\} \\ &= \sum_{m=0}^k (-1)^m C(k, m) y_{k-m} \text{Av}\{\varepsilon_{k-m}\} = 0 \\ \text{Var}\{\Delta^k y_i \varepsilon_i\} &= \text{Av}\left\{ \sum_{m=0}^k (-1)^m C(k, m) y_{k-m} \varepsilon_{k-m} \right\}^2 \\ &= \sum_{r=0}^k \sum_{s=0}^k (-1)^{r+s} C(k, r) C(k, s) y_{k-r} y_{k-s} \text{Av}\{\varepsilon_{k-r} \varepsilon_{k-s}\} \end{aligned}$$

Only when  $r = s$  is the average not zero, hence

$$\begin{aligned}\text{Var}\{\Delta^k y_i \varepsilon_i\} &= \sum_{r=0}^k C^2(k, r) \text{Av}\{y_{k-r} \varepsilon_{k-r}\}^2 \\ &= \sum C^2(k, r) \sigma^2\end{aligned}$$

which defines  $\sigma^2$  as the original “noise.” Using the identity of Prob. 10.4.1,

$$\text{Var}\{\Delta^k y_i \varepsilon_0\} = C(2k, k) \sigma^2 = \frac{(2k)!}{(k!)^2} \sigma^2$$

The result we have is that the variance in the  $k$ th difference column is  $C(2k, k)$  times the variance in the original table values—differencing *amplifies* the noise in a table by a factor  $(2k)!/(k!)^2$ . (See Table 10.4.)

Therefore, in order to estimate the variance (noise) in a table, we estimate the noise by finding the noise in the  $k$ th difference column and dividing this by  $C(2k, k)$ .

Sections 10.5 and 10.6 will show which value of  $k$  to use.

Table 10.4 ROUNDOFF AMPLIFICATION FACTORS

| Order of difference | Variance $C(2k, k)$ | Root-mean-square noise $\sqrt{C(2k, k)}$ | Maximum noise $2^k$ |
|---------------------|---------------------|------------------------------------------|---------------------|
| 1                   | 2                   | 1.414                                    | 2                   |
| 2                   | 6                   | 2.449                                    | 4                   |
| 3                   | 20                  | 4.472                                    | 8                   |
| 4                   | 70                  | 8.367                                    | 16                  |
| 5                   | 252                 | 15.875                                   | 32                  |
| 6                   | 924                 | 30.397                                   | 64                  |

## PROBLEMS

10.4.1 Expand both sides of  $(1+t)^{a+b} = (1+t)^a(1+t)^b$  and equate like powers of  $t$  to get

$$C(a+b, r) = \sum_{s=0}^a C(a, s) C(b, r-s)$$

10.4.2 In Prob. 10.4.1 set  $a = b = r$  to get

$$C(2r, r) = \sum_{s=0}^r C^2(r, s)$$

10.4.3 In Prob. 10.4.1, set  $a = b = n$  and  $r = n+1$  to get

$$C(2n, n+1) = \sum C(n, s) C(n, s-1)$$

### 10.5 CORRELATION IN THE $k$ TH DIFFERENCES

The consecutive values in a column of  $k$ th differences of a difference table use many of the same numbers, but an examination of how they enter shows that they have opposite signs in the consecutive differences—they are negatively correlated. We need to compute this correlation

$$\begin{aligned} \text{Av} \left\{ \sum_{r=0}^k \frac{(-1)^r C(k, r) y_{k-r} \varepsilon_{k-r}}{\sqrt{C(2k, k)}} \cdot \sum_{r=1}^{k+1} \frac{(-1)^{r-1} C(k, r-1) y_{k-r} \varepsilon_{k-r}}{\sqrt{C(2k, k)}} \right\} \\ = - \frac{1}{C(2k, k)} \sum_{r=0}^k C(k, r) C(k, r-1) \sigma^2 \\ = - \frac{C(2k, k+1)}{C(2k, k)} \sigma^2 = \frac{-k}{k+1} \sigma^2 \end{aligned}$$

where we have used the result from Prob. 10.4.3.

As we expected, the coefficient of correlation of the noise from successive entries in the  $k$ th column of a difference table is large and negative,  $-k/(k+1)$ . This is well known in the folklore of computing, but it is rarely estimated by theoretical considerations.

If the roundoff values are drawn from a normal Gaussian distribution, then these correlation coefficients imply a probability of a sign change indicated in Table 10.5.

This alternation of sign will give us a way of estimating when the difference table column is noise rather than signal.

This is a theoretical result in the sense that we have averaged over the imagined ensemble of similar tables, and in practice all we have is the single table. The

Table 10.5\*

| $k$ | $-k/k+1$          | Probability,<br>percentage |
|-----|-------------------|----------------------------|
| 0   | 0 = 0             | .500                       |
| 1   | $-1/2 = -0.500$   | .668                       |
| 2   | $-2/3 = -0.667$   | .733                       |
| 3   | $-3/4 = -0.750$   | .770                       |
| 4   | $-4/5 = -0.800$   | .796                       |
| 5   | $-5/6 = -0.834$   | .814                       |
| 6   | $-6/7 = -0.857$   | .828                       |
| 7   | $-7/8 = -0.875$   | .839                       |
| 8   | $-8/9 = -0.890$   | .850                       |
| 9   | $-9/10 = -0.900$  | .857                       |
| 10  | $-10/11 = -0.910$ | .864                       |
| 20  | $-20/21 = -0.952$ | .902                       |

\* From Roger Pinkham.

solution to this dilemma is the standard one in such situations; we appeal to the *ergodic hypothesis* which states that (for suitable situations) “the average over the ensemble is the same as the average over the function.” A simple illustration of this is the very common habit of looking at, say, a table of 100,000 deaths and applying this to a single life. Or again, looking at the distribution of salaries paid (as of the current year) as a function of age and seeing the individual progress along the chart as he ages.

The ergodic hypothesis can be applied when the statistics are stationary (not changing in time or from place to place). In practice we rarely have perfectly stationary situations, and we must make judgments as to how much the results could be altered by the changes. In the two examples above, we sense that the health of the general population is not changing so rapidly that the table of 100,000 deaths is totally irrelevant, and the effect of the slow changes can be mentally added on. In the salary curve example, however, inflation and the changing status of jobs and job titles all make the curves dangerous to use as a basis for applying the ergodic hypothesis. Similarly, in tables and other computations we do not expect to find that all the results have the same level of roundoff noise. We can apply the hypothesis only to runs where the statistics do not seem to be changing too rapidly. In particular, the interval we use of the difference column should have numbers about the same size and about the same density as the sign changes.

## 10.6 ESTIMATION OF ROUNDOFF IN A TABLE

We have now examined all the parts needed in a theory of the estimation of roundoff in a table of results computed for equally spaced values of a variable or a parameter. The importance of being able to make an estimate of the random error in a table of results *from the results alone* should be recognized; it is not a perfect answer, but it is better than nothing.

First we have shown that the  $(N + 1)$ st differences of a polynomial are zero, and from experience we know that many functions are locally very like a polynomial. Thus we expect that the differences of many functions will approach zero as more and more differences are taken.

If the differences do not become small fairly rapidly as  $k$  becomes large, then it is likely that the spacing of the values is too large. It is easy to see that if the spacing is halved, then the first differences are about half as large as they were, the second differences one-quarter as large, the third, one-eighth, and in general the  $k$ th differences will be about  $1/2^k$  times as large as they were. Thus it is generally true that, except near a singularity, for a smooth function the differences approach zero as  $k$  gets large.

On the other hand, we have just seen that random noise in a table rapidly approaches infinity and that the successive values in a particular column tend strongly to alternate in sign. Given a table of computed results, we expect a mixture of the two effects, the true differences approaching zero and the noise rapidly growing and alternating in sign.

In most reasonably planned computations the fourth or fifth differences begin to show the oscillation we identify with the noise. Very smooth computations, such as orbits of the planets in space, may have reasonably smooth differences up to at least the tenth order, and very rough computations may have the second or third difference showing alternating signs; but typically, as we said, it is at about the fourth or fifth difference that oscillation occurs. We will, for the sake of simplicity, suppose that we are interested in the fourth or fifth difference.

We need a specific test for a machine to make. Just what we use depends on how certain we wish to be, but as a general rule if we find two out of three or, safer, three out of four sign changes in consecutive values of a difference column, then we will call that the noise.

To estimate the noise we compute the average of squares (the variance) of this difference column, using all the values appearing to have somewhat the same oscillation rate and size. Using the ergodic hypothesis, we equate this number to the average over the imagined ensemble of tables. To get from the noise in the  $k$ th differences back to the noise in the table we divide by  $C(2k, k) = (2k)!/(k!)^2$ . This gives the square of the average noise in the original table.

Table 10.6.1

| $x$ | $f(x)$ | $\Delta$ | $\Delta^2$ | $\Delta^3$ | $\Delta^4$ |
|-----|--------|----------|------------|------------|------------|
| 1.0 | 1.000  |          |            |            |            |
| 1.1 | 0.951  | -49      |            |            |            |
|     |        | -33      | +16        |            |            |
| 1.2 | 0.918  | -21      | +12        | -4         | 3          |
|     |        | -10      | +11        | -1         |            |
| 1.3 | 0.897  | -1       | +9         | -2         | -1         |
|     |        | +8       | +9         | 0          | +2         |
| 1.4 | 0.887  | +15      | +7         | -2         | -2         |
|     |        | +22      | +7         | 0          | +2         |
| 1.5 | 0.886  | +31      | +9         | +2         | +2         |
|     |        |          |            |            |            |
| 1.6 | 0.894  |          |            |            |            |
| 1.7 | 0.909  |          |            |            |            |
| 1.8 | 0.931  |          |            |            |            |
| 1.9 | 0.962  |          |            |            |            |

As an example consider Table 10.6.1 of the gamma function. The average of the squares of the fourth differences is

$$\text{Var}\{\Delta^4\} = \frac{9 + 1 + 4 + 4 + 4 + 4}{6} = \frac{26}{6} = \frac{13}{3}$$

We divide this by  $C(8, 4) = 70$  to get  $0.062 = \sigma^2$  which is an estimate of the noise in the table.

The table supposedly has only random roundoff in the last digit, and the corresponding model for the flat distribution is

$$\sigma_2^2 = \int_{-1/2}^{1/2} x^2 dx = \frac{1}{12} = 0.0833.$$

Thus we have the three values

$$\sigma_1 = 0.25$$

$$\sigma_2 = 0.29$$

$$\sigma_3 = 0.32$$

Table 10.6.2

| $x$ | $\zeta(x)$ | $\Delta\zeta$ | $\Delta^2\zeta$ | $\Delta^3\zeta$ | $\Delta^4\zeta$ | $\Delta^5\zeta$ |
|-----|------------|---------------|-----------------|-----------------|-----------------|-----------------|
| 2   | 1.64493    |               |                 |                 |                 |                 |
|     |            | -0.44287      |                 |                 |                 |                 |
| 3   | 1.20206    |               | 0.32313         |                 |                 |                 |
|     |            | -0.11974      |                 | -0.24878        |                 |                 |
| 4   | 1.08232    |               | 7435            |                 | 0.20023         |                 |
|     |            | -0.04539      |                 | -4855           |                 | -0.16688        |
| 5   | 1.03693    |               | 2580            |                 | 3335            |                 |
|     |            | -0.01959      |                 | -1520           |                 | -2403           |
| 6   | 1.01734    |               | 1060            |                 | 932             |                 |
|     |            | -0.00899      |                 | -588            |                 | -596            |
| 7   | 1.00835    |               | 472             |                 | 336             |                 |
|     |            | -0.00427      |                 | -252            |                 | -199            |
| 8   | 1.00408    |               | 220             |                 | 137             |                 |
|     |            | -0.00207      |                 | -115            |                 | -75             |
| 9   | 1.00201    |               | 105             |                 | 62              |                 |
|     |            | -0.00102      |                 | -53             |                 | -35             |
| 10  | 1.00099    |               | 52              |                 | 27              |                 |
|     |            | -0.00050      |                 | -26             |                 | -16             |
| 11  | 1.00049    |               | 26              |                 | 11              |                 |
|     |            | -0.00024      |                 | -15             |                 | -0              |
| 12  | 1.00025    |               | 11              |                 | 11              |                 |
|     |            | -0.00013      |                 | -4              |                 | -11             |
| 13  | 1.00012    |               | 7               |                 | 0               |                 |
|     |            | -0.00006      |                 | -4              |                 | +3              |
| 14  | 1.00006    |               | 3               |                 | 3               |                 |
|     |            | -0.00003      |                 | -1              |                 | -4              |
| 15  | 1.00003    |               | 2               |                 | -1              |                 |
|     |            | -0.00001      |                 | -2              |                 |                 |
| 16  | 1.00002    |               | 0               |                 |                 |                 |
|     |            | -0.00001      |                 |                 |                 |                 |
| 17  | 1.00001    |               |                 |                 |                 |                 |

where  $\sigma_3$  is from a careful, exact computation of the roundoffs (knowing the correct values from a much more accurate table). In view of the smallness of our sample the estimate is fairly good.

We observed that the theory need not apply near a singularity. The zeta function

$$\zeta(x) = \sum_{k=1}^{\infty} \frac{1}{k^x}$$

clearly is infinite at  $x = 1$  and has a singularity there. Table 10.6.2 shows how the differences go.

We see signs of roundoff noise at the bottom of the fifth difference column, but at the top we see the effect of the singularity.

On the other hand, a table of a function like the error integral shows that the effects we have claimed usually happen (see Table 10.6.3)

Table 10.6.3

| $t$  | $f(t)$ | $\Delta$ | $\Delta^2$ | $\Delta^3$ | $\Delta^4$ | $\Delta^5$ |
|------|--------|----------|------------|------------|------------|------------|
| 0.00 | 0.0000 |          |            |            |            |            |
|      |        | 987      |            |            |            |            |
| 0.25 | 0.0987 |          | -59        |            |            |            |
|      |        | 928      |            | -50        |            |            |
| 0.50 | 0.1915 |          | -109       |            | +19        |            |
|      |        | 819      |            | -31        |            | +4         |
| 0.75 | 0.2734 |          | -140       |            | +23        |            |
|      |        | 679      |            | -8         |            | -10        |
| 1.00 | 0.3413 |          | -148       |            | +13        |            |
|      |        | 531      |            | +5         |            | +4         |
| 1.25 | 0.3944 |          | -143       |            | +17        |            |
|      |        | 388      |            | +22        |            | -11        |
| 1.50 | 0.4332 |          | -121       |            | +6         |            |
|      |        | 267      |            | +28        |            | -10        |
| 1.75 | 0.4599 |          | -93        |            | -4         |            |
|      |        | 174      |            | +24        |            | +4         |
| 2.00 | 0.4773 |          | -69        |            | 0          |            |
|      |        | 105      |            | +24        |            | -7         |
| 2.25 | 0.4878 |          | -45        |            | -7         |            |
|      |        | 60       |            | +17        |            | +3         |
| 2.50 | 0.4938 |          | -28        |            | -4         |            |
|      |        | 32       |            | +13        |            | -4         |
| 2.75 | 0.4970 |          | -15        |            | -8         |            |
|      |        | 17       |            | +5         |            | +10        |
| 3.00 | 0.4987 |          | -10        |            | +2         |            |
|      |        | 7        |            | +7         |            | -9         |
| 3.25 | 0.4994 |          | -3         |            | -7         |            |
|      |        | 4        |            | 0          |            | +9         |
| 3.50 | 0.4998 |          | -3         |            | +2         |            |
|      |        | 1        |            | +2         |            |            |
| 3.75 | 0.4999 |          | -1         |            |            |            |
|      |        | 1        |            |            |            |            |
| 4.00 | 0.5000 |          |            |            |            |            |



### PROBLEMS

- 10.6.1 Estimate the noise in Table 10.6.3.  
 10.6.2 Carry out the differences in Table 10.6.2 to order 7 and use the last few values to make a "desperate" estimate of the noise.  
 10.6.3 Discuss using the ergodic hypothesis to estimate the total high school population from the fraction of one's life that is spent in high school. Discuss the effects of nonstationarity in the population.  
 10.6.4 Discuss the roundoff in Table 10.5 of correlation coefficient probabilities.

### 10.7 ISOLATED ERRORS

It was very common in hand calculation to have an isolated error in a table; it is less common but by no means impossible in modern machine computation. The isolated error may well be due to the fact that for that particular value an indeterminate form arose and led to large roundoff errors.

It is very easy to locate the error by looking at the difference table. We know from Table 10.3 that this leads to a series of terms of the error times the binomial coefficients with alternating signs. For example, in reviewing Table 10.7, the author found an error by this simple device of looking at the difference table.

The pattern suggested placing the fourth differences opposite the value of 35°. Using the middle three values

$$\begin{aligned} 41 &= -4\varepsilon + C \\ -59 &= 6\varepsilon + C \\ 37 &= -4\varepsilon + C \end{aligned}$$

we solve the equations approximately. They suggest  $\varepsilon = -10$ . If we change the value at 35° to 6,421, we will have the corrections to the difference table

|       |      |     |   |    |
|-------|------|-----|---|----|
|       |      |     |   | —4 |
|       |      |     | 8 |    |
|       |      | —50 | 1 |    |
|       | —468 | 9   |   |    |
| 6,421 | —41  | 1   |   |    |
|       | —509 | 10  |   |    |
|       | —31  | —3  |   |    |
|       |      | 7   |   |    |
|       |      |     | 0 |    |

and this looks much better (and the error represents a typical typographical error).

Occasionally it is necessary to supply a missing value in a table of equally spaced values. For this we go back to the differences written out in terms of the original function values, Sec. 10.3. If we pick an even-order difference, say 6, and suppose that this order of differences is zero, then we have, assuming for convenience that the error is at zero,

$$y_{-3} - 6y_{-2} + 15y_{-1} - 20y_0 + 15y_1 - 6y_2 + y_3 = 0$$

or

$$y_0 = \frac{1}{20}[(y_3 + y_{-3}) - 6(y_2 + y_{-2}) + 15(y_1 + y_{-1})]$$

Table 10.7

| Degree | $C_n(z)$ | $\Delta$ | $\Delta^2$ | $\Delta^3$ | $\Delta^4$ | Theoretical<br>fourth<br>difference<br>due to error $\epsilon$ |
|--------|----------|----------|------------|------------|------------|----------------------------------------------------------------|
| 0      | 8,346    |          |            |            |            |                                                                |
|        |          | -44      |            |            |            |                                                                |
| 5      | 8,302    | -131     | -87        |            |            |                                                                |
| 10     | 8,171    | -214     | -83        | 4          |            |                                                                |
| 15     | 7,957    | -290     | -76        | 7          | 3          |                                                                |
| 20     | 7,667    | -360     | -70        | 6          | -1         |                                                                |
| 25     | 7,307    | -418     | -58        | 12         | 6          |                                                                |
| 30     | 6,889    | -478     | -60        | -2         | -14        | $+\epsilon$                                                    |
| 35     | 6,411    | -499     | -21        | 39         | 41         | $-4\epsilon$                                                   |
| 40     | 5,912    | -540     | -41        | -20        | -59        | $+6\epsilon$                                                   |
| 45     | 5,372    | -564     | -24        | 17         | 37         | $-4\epsilon$                                                   |
| 50     | 4,808    | -581     | -17        | 7          | -10        | $+\epsilon$                                                    |
| 55     | 4,227    | -593     | -12        | 5          | -2         |                                                                |
| 60     | 3,634    | -600     | -7         | 5          | 0          |                                                                |
| 65     | 3,034    | -604     | -4         | 3          | -2         |                                                                |
| 70     | 2,430    | -607     | -3         | 1          | -2         |                                                                |
| 75     | 1,823    | -607     | -0         | 3          | 2          |                                                                |
| 80     | 1,216    | -607     | -1         | -1         | -4         |                                                                |
| 85     | 608      | -608     | 0          | 1          | 2          |                                                                |
| 90     | 0        |          |            |            |            |                                                                |

### PROBLEMS

- 10.7.1 Estimate the noise in the corrected table in this section.
- 10.7.2 Correct Table 10.7 by regarding the values of  $35^\circ$  as missing.
- 10.7.3 Discuss the dangers of trying to correct a bad value because the formula at that point was indeterminate.

### 10.8 SYSTEMATIC ERRORS

We have provided a method of estimating the random errors in a table. The method does nothing about systematic errors. One way to check an important table is to compute, by some *independent* method, several key values scattered across the range of values. This is easy to say (and hard to do in many cases) but without some redundant, independent calculations there can be no realistic checking of the results. The fact that the theoretician accepts the results is no measure of their correctness because, in a very real sense, a good theoretician can account for any new phenomenon he meets—it is his business to create new theories to fit new data!

The situation is the same as in experimental work. It is possible to estimate, from the measurements alone, the random component of the error, but the calculation of the systematic errors is another thing entirely and requires other methods.<sup>1</sup> The fact that it is hard is no reason for ignoring the problem. Any numerical analyst must, if he is a responsible scientist and not a hack technician, assume responsibility for the accuracy of his results and some of the burden for the correctness of the original formulation, as well as for the interpretation of the results. The job of a numerical analyst neither begins with the given equations nor ends with the output sheet. Realistic estimates of the total error in the answer are partly his responsibility, not exclusively that of the problem's proposer.

<sup>1</sup> W. J. Youden, Enduring Values, *Technometrics*, vol. 14, no. 1, February 1972, pp. 1–11.

## 11.1 INTRODUCTION

The difference and integral calculi are related to each other by two formulas

$$f(x) = \frac{d}{dx} \int_a^x f(x) dx$$

$$f(x) - f(a) = \int_a^x \frac{df}{dx} dx$$

Similarly, the difference and summation calculi are related by

$$f(n+1) = \Delta \sum_{k=0}^n f(k) = \sum_{k=0}^{n+1} f(k) - \sum_{k=0}^n f(k)$$

$$f(n+1) - f(0) = \sum_{k=0}^n \Delta f(k) = \sum_{k=0}^n [f(k+1) - f(k)]$$

where we have used the common summation notation

$$\sum_{k=a}^n f(k) = f(a) + f(a+1) + \cdots + f(n) \quad \sum_{k=a}^a f(a) = f(a) \quad \sum_{k=a}^{a-1} f(a) = 0$$

Because of the arguments  $n + 1$  in the two formulas, Boole [4], Jordan [25], and others have used the notation for the sum

$$\sum_{k=a}^n f(k) = f(a) + f(a+1) + \cdots + f(n-1) \quad \sum_{k=a}^a f(k) = 0$$

However, this differs from the common usage in mathematics, and we shall therefore avoid laying that trap for the unwary and instead accept the annoyance of the  $n + 1$  that occurs in some of the arguments. J. W. Tukey<sup>1</sup> has suggested the elegant notation

$$\sum_{k=a}^{<b} f(k) = f(a) + f(a+1) + \cdots + f(b-1)$$

as the solution to this dilemma, but we will not use it here.

In the integral calculus a table of derivatives of various functions is inverted to get the basic table of integrals. For example, from

$$\frac{d}{dx} x^n = nx^{n-1}$$

we get

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

In a similar way, from the table of first differences of various functions we get (using unit spacing in the differences)

$$\begin{aligned} \sum_{x=0}^m x^{(n)} &= \frac{(m+1)^{(n+1)}}{n+1} & n \neq -1 \\ \sum_{x=0}^m a^x &= \frac{a^{m+1} - 1}{a - 1} & a \neq 1 \\ \sum_{x=1}^m \ln x &= \ln(m!) \\ \sum_{x=0}^m \sin(ax + b) &= -\frac{\cos[a(m + \frac{1}{2}) + b] - \cos(-a/2 + b)}{2 \sin(a/2)} \\ &= \frac{\sin[a(m+1)/2] \sin(am/2 + b)}{\sin(a/2)} \\ \sum_{x=0}^m \cos(ax + b) &= \frac{\sin[a(m + \frac{1}{2}) + b] - \sin(-a/2 + b)}{2 \sin(a/2)} \\ &= \frac{\sin[a(m+1)/2] \cos(am/2 + b)}{\sin(a/2)} \end{aligned}$$

Many others are easily found, but we do not need them now.

<sup>1</sup> J. W. Tukey (1915—).

## PROBLEMS

11.1.1 Compute the sum  $1 + 2 + 4 + \cdots + 2^{64}$ .

11.1.2 Show that

$$\sum_{x=0}^{2N-1} \cos \frac{\pi x}{N} = 0 \quad \sum_{x=0}^{2N-1} \sin \frac{\pi x}{N} = 0$$

11.1.3 Using Prob. 11.1.2, show that

$$\sum_{x=0}^{2N-1} \cos \frac{\pi k x}{N} \cos \frac{\pi m x}{N} = N \delta(m, k) \quad 0 < m + k < 2N$$

$$\sum_{x=0}^{2N-1} \cos \frac{\pi k x}{N} \sin \frac{\pi m x}{N} = 0$$

$$\sum_{x=0}^{2N-1} \sin \frac{\pi k x}{N} \sin \frac{\pi m x}{N} = N \delta(m, k) \quad 0 < m + k < 2N$$

11.1.4 Do the same as in Prob. 11.1.3 except use  $N - 1$  as the upper limit and  $2\pi$  in place of  $\pi$ .

## 11.2 SUMMATION BY PARTS

An examination of the general processes used in the calculus of analytic integration reveals that there are two principal tools:

- 1 Change of variables
- 2 Integration by parts

Of these two methods, the first is not available in the summation calculus, which depends on equally spaced arguments, where usually  $h = 1$ . This makes computing a sum in an analytic, closed form much more difficult than computing an integral.

On the other hand the analog of integration by parts is summation by parts, and it plays the corresponding role in its field. Integration by parts in the calculus is based on the derivative of a product

$$d(uv) = u dv + v du$$

From the above equation the formula for integration by parts

$$\int u dv = uv - \int v du$$

is found by integration. Similarly, from the formula for the difference of a product

$$\Delta u(x)v(x) = u(x) \Delta v(x) + v(x+1) \Delta u(x)$$

we obtain by summation

$$\sum_{x=0}^{m-1} u \Delta v = u(m)v(m) - u(0)v(0) - \sum_{x=0}^{m-1} v(x+1) \Delta u(x)$$

We can choose  $v(x)$  so that  $v(0)$  equals zero (or any other value we please); just as in the calculus the constant of integration in finding the  $v$  can be any number we please (conventionally it is picked to be zero).

As an example of the use of summation by parts, consider

$$\begin{aligned}\sum_{x=0}^{m-1} xa^x &= \left. \frac{xa^x}{a-1} \right|_0^m - \sum_0^{m-1} \frac{a^{x+1}}{a-1} \cdot 1 \\ &= \frac{ma^m}{a-1} - \frac{a^{m+1}}{(a-1)^2} \Big|_0^m = \frac{a}{(1-a)^2} [(m-1)a^m - ma^{m-1} + 1] \quad a \neq 1\end{aligned}$$

In general, the use of summation by parts closely resembles that of integration by parts. For example, summations such as

$$\sum \frac{1}{2^x} \sin \theta x$$

are accomplished by using summation by parts twice and then combining like terms. It can also be done by writing it as

$$\left\{ \left( \frac{e^{i\theta}}{2} \right)^x - \left( \frac{e^{-i\theta}}{2} \right)^x \right\} \frac{1}{2i}$$

and summing directly.

## PROBLEMS

11.2.1 Show that

$$\sum_{x=1}^n \frac{1}{2^x} \sin \theta x = \frac{2 \sin \theta - 2^{-n} [2 \sin \theta(n+1) - \sin \theta]}{1 + 8 \sin^2(\theta/2)}$$

11.2.2 Sum  $\sum_{x=1}^n x^2 \cos x$ .

11.2.3 Sum  $\sum_{x=1}^n x \ln x$ .

11.2.4 Sum  $\sum_{x=0}^n x^2 \cdot 2^x$ .

11.2.5 Sum  $\sum_{x=0}^n x^2 \cdot x^{(2)}$ .

## 11.3 SUMMATION OF POWERS OF $x$

The special case of summing the powers of the consecutive integers is of enough importance to justify a special examination. We could, of course, sum them by converting the power of  $x$  via the Stirling numbers of the second kind (Sec. 9.8) and summing the result by the obvious formula.

Instead, we shall approach the problem much as Jacob Bernoulli [3] did, and in the process we will introduce two valuable by-products: (1) the idea of a generating function, and (2) the Bernoulli numbers. Both will have further use.

The summation of the first few cases of the powers of  $x$  is easily remembered if written in the form

$$\begin{aligned}
 \sum_{x=1}^n x &= \sum_1^n x^{(1)} = \frac{(n+1)n}{2} \\
 \sum_{x=1}^n x^2 &= \sum_1^n (x^{(2)} + x^{(1)}) = \frac{(n+1)n(2n+1)}{6} \\
 \sum_{x=1}^n x^3 &= \sum_1^n (x^{(3)} + 3x^{(2)} + x^{(1)}) = \left[ \frac{(n+1)n}{2} \right]^2 \\
 \sum_{x=1}^n x^4 &= \frac{(n+1)n(2n+1)}{6} \frac{3n^2 + 3n - 1}{5} \\
 \sum_{x=1}^n x^5 &= \left[ \frac{(n+1)n}{2} \right]^2 \frac{2n^2 + 2n - 1}{3} \\
 \sum_{x=1}^n x^6 &= \frac{(n+1)n(2n+1)}{6} \frac{3n^4 + 6n^3 - 3n + 1}{7} \\
 \sum_{x=1}^n x^7 &= \left[ \frac{(n+1)n}{2} \right]^2 \frac{3n^4 + 6n^3 - n^2 - 4n + 2}{6}
 \end{aligned}$$

The summation formula for  $x^{(n)}$  is also valid for negative exponents ( $n \neq -1$ ). For example,

$$\begin{aligned}
 \sum_{x=1}^m \frac{1}{x(x+1)} &= \sum_{x=1}^m (x-1)^{(-2)} = \frac{m^{(-1)} - 0^{(-1)}}{-1} \\
 &= -\frac{1}{m+1} + 1 = \frac{m}{m+1}
 \end{aligned}$$

Similarly,

$$\begin{aligned}
 \sum_1^m \frac{1}{x(x+1)(x+2)} &= \sum_1^m (x-1)^{(-3)} = \frac{m^{(-2)} - 0^{(-2)}}{-2} \\
 &= \frac{1}{2} \left[ \frac{1}{1 \cdot 2} - \frac{1}{(m+1)(m+2)} \right]
 \end{aligned}$$



## PROBLEMS

11.3.1 Verify the formulas for  $x^3$  and  $x^5$ , using the synthetic-division method.

11.3.2 Using  $\Delta C(n, k) = C(n+1, k) - C(n, k) = C(n, k-1)$ , show that

$$\sum_{x=k}^m C(x, k) = C(m+1, k+1)$$

11.3.3 Show that

$$\begin{aligned}\sum_{x=1}^n (2x-1) &= n^2 \\ \sum_{x=1}^n (2x-1)^2 &= \frac{n(4n^2-1)}{3} \\ \sum_{x=1}^n (2x-1)^3 &= n^2(2n^2-1)\end{aligned}$$

11.3.4 Compute

$$\sum_{x=1}^n \frac{1}{x(x+1)(x+2)(x+3)}$$

11.3.5 Compute

$$\sum_{m=0}^{N-1} \sin^m \theta \quad \theta \neq 0, \pm\pi, \dots$$

Ans.  $(1 - \sin^N \theta)/(1 - \sin \theta)$

## 11.4 GENERATING FUNCTIONS

The idea of a **generating function** is simple once it is well understood, but initially it seems to be baffling. We will therefore begin by a few simple examples.

The binomial coefficients of a given order  $n$

$$C(n, k) \quad k = 0, 1, 2, \dots, n$$

can be multiplied by the corresponding  $t^k$  and written as a sum. The binomial theorem shows that this sum is easily written as

$$(1+t)^n = C(n, 0) + C(n, 1)t + C(n, 2)t^2 + \dots + C(n, n)t^n$$

The expression  $(1+t)^n$  is said to be a *generating function* for the binomial coefficients.

The power of generating functions lies in the ease of manipulating them in certain processes. For example, again using the binomial coefficients, if we multiply a generating function of order  $m$  by a second of order  $n$ , we get the generating function of order  $m+n$ ; that is,

$$(1+t)^m(1+t)^n = (1+t)^{m+n}$$

Thus we have

$$\sum_{k=0}^m C(m, k)t^k \sum_{j=0}^n C(n, j)t^j = \sum_{i=0}^{m+n} C(m+n, i)t^i$$

Equating like powers of  $t$ , we get, as in Prob. 10.4.1,  $m+n+1$  identities.

$$\sum_{k=0}^m C(m, k)C(n, i-k) = C(m+n, i) \quad i = 0, 1, \dots, m+n$$

Consider next the Stirling numbers of the first kind

$$x^{(n)} = \sum_{k=0}^n s(n, k)x^k$$

as another example of a generating function for a set of numbers

$$s(n, k) \quad k = 0, 1, \dots, n$$

with  $x$  as the “dummy variable.”

Instead of using only powers of  $t$  as the place holders we may use  $t^k/k!$  if we please, or any other multiple of  $t^k$ . That is why we used the words *a generating function for*, though usually *the generating function* means just  $t^k$ . The use of  $t^k/k!$  is often called “the exponential generating function.”

## PROBLEMS

11.4.1 Show that

$$\sum_{k=0}^n (-1)^k C(n, k) = 0$$

11.4.2 Show that

$$\sum_{k=0}^n C(n, k) = 2^n$$

11.4.3 In the generating function for the Stirling numbers of the first kind set the variable equal to 1 to obtain

$$\sum_{k=0}^n s(n, k) = 0 \quad n \geq 1$$

## 11.5 SUMS OF POWERS OF $n$ AGAIN

We propose to find the sums [where  $S(n, p)$  is *not* a Stirling number]

$$S(n, p) = 1^p + 2^p + 3^p + \dots + n^p = \sum_{k=1}^n k^p$$

using the method of generating functions. In this method we multiply  $S(n, p)$  by  $t^p/p!$  and sum over all  $p$  to get

$$\begin{aligned}\sum_{p=0}^{\infty} \frac{S(n, p)t^p}{p!} &= \sum_{p=0}^{\infty} \sum_{k=1}^n \frac{k^p t^p}{p!} \\ &= \sum_{k=1}^n \sum_{p=0}^{\infty} \frac{(kt)^p}{p!} = \sum_{k=1}^n e^{kt} \\ &= \frac{e^{(n+1)t} - e^t}{e^t - 1}\end{aligned}$$

Thus we have the exponential generating function for the sums.

To find the coefficient of  $t^p/p!$ , we write

$$\frac{e^{(n+1)t} - e^t}{e^t - 1} = \frac{e^{nt} - 1}{t} \frac{te^t}{e^t - 1}$$

The first factor can be written in the form

$$\frac{e^{nt} - 1}{t} = \frac{\sum_{k=0}^{\infty} (nt)^k/k! - 1}{t} = \sum_{k=0}^{\infty} \frac{n^{k+1}t^k}{(k+1)!}$$

The second factor requires more attention. We begin with the expansion (see Sec. 11.6 for details)

$$\frac{t}{e^t - 1} = \sum_{k=0}^{\infty} \frac{B_k t^k}{k!}$$

Solving for the first few coefficients, we find  $B_0 = 1$  and  $B_1 = -1/2$ , so that

$$\frac{t}{e^t - 1} = 1 - t/2 + \sum_{k=2}^{\infty} \frac{B_k t^k}{k!}$$

Transposing the  $t/2$  term, we have

$$t \left( \frac{1}{e^t - 1} + \frac{1}{2} \right) = 1 + \sum_{k=2}^{\infty} \frac{B_k t^k}{k!}$$

If we replace  $t$  by  $-t$  on the left-hand side, we get

$$\begin{aligned}-t \left[ \frac{1}{e^{-t} - 1} + \frac{1}{2} \right] &= -\frac{t}{2} \frac{1 + e^{-t}}{e^{-t} - 1} = \frac{t}{2} \frac{e^t + 1}{e^t - 1} \\ &= \frac{t}{e^t - 1} + \frac{t}{2}\end{aligned}$$

which is the original expression. Thus, except for  $B_1$  which equals  $-1/2$ , all the other  $B_{2k+1} = 0$  ( $k = 1, 2, \dots$ ).

From this we see that the second factor

$$\begin{aligned}\frac{te^t}{e^t-1} &= t + \frac{t}{e^t-1} \\ &= t + 1 - \frac{t}{2} + \sum_{m=1}^{\infty} \frac{B_{2m} t^{2m}}{(2m)!} \\ &= 1 + \frac{t}{2} + \sum_{m=1}^{\infty} \frac{B_{2m} t^{2m}}{(2m)!}\end{aligned}$$

To find the expansion of the product of the two factors, we multiply the two series together and pick out the coefficient of  $t^p/p!$

$$\begin{aligned}p! \left[ \frac{n^{p+1}}{(p+1)!} + \frac{n^p}{p!2} + \frac{n^{p-1}B_2}{(p-1)!2!} + \frac{n^{p-3}B_4}{(p-3)!4!} + \cdots \right] \\ = \frac{n^{p+1}}{p+1} + \frac{n^p}{2} + \frac{p}{2} B_2 n^{p-1} + \frac{p(p-1)(p-2)}{2 \cdot 3 \cdot 4} B_4 n^{p-3} \\ + \frac{p(p-1)(p-2)(p-3)(p-4)}{2 \cdot 3 \cdot 4 \cdot 5 \cdot 6} B_6 n^{p-5} + \cdots\end{aligned}$$

which is what Bernoulli obtained [3, p. 97].

## PROBLEMS

11.5.1 We can write the generating function as

$$\begin{aligned}\frac{e^{(n+1)t} - e^{-t}}{e^t - 1} &= \frac{e^{(n+1)t} - 1}{e^t - 1} - 1 \\ &= \frac{e^{(n+1)t} - 1}{t} \frac{t}{e^t - 1} - 1\end{aligned}$$

and get an expansion for  $S(n, p)$  in powers of  $n+1$ . Carry out the details.

11.5.2 Show that the formulas in Sec. 11.3 follow from

$$B_0 = 1 \quad B_1 = -1/2 \quad B_2 = 1/6 \quad B_4 = -1/30 \quad B_6 = 1/42.$$

## 11.6 THE BERNOULLI NUMBERS

The generating function for the Bernoulli numbers is

$$\frac{t}{e^t - 1} = \sum_{k=0}^{\infty} \frac{B_k t^k}{k!}$$

We have already found that

$$B_0 = 1 \quad B_1 = -1/2 \quad B_{2k+1} = 0 \quad k = 1, 2, \dots$$

To investigate these numbers further we write the first equation as

$$\begin{aligned} t &= (e^t - 1) \sum_{k=0}^{\infty} \frac{B_k t^k}{k!} \\ &= \sum_{m=1}^{\infty} \frac{t^m}{m!} \sum_{k=0}^{\infty} \frac{B_k t^k}{k!} \\ &= \sum_{p=1}^{\infty} t^p \sum_{s=0}^{p-1} \frac{B_s}{s!(p-s)!} \end{aligned}$$

For  $p = 1$  we have, equating like powers of  $t$ ,

$$t = t \cdot B_0 \rightarrow B_0 = 1$$

For  $p > 1$

$$0 = \sum_{s=0}^{p-1} \frac{B_s}{s!(p-s)!}$$

Multiply by  $p!$

$$0 = \sum_{s=0}^{p-1} \frac{p!}{s!(p-s)!} B_s = \sum_{s=0}^{p-1} C(p, s) B_s$$

This formula is easily remembered by the mnemonic of writing it as

$$(B + 1)^p - B^p = 0 \quad p > 1$$

and then writing each  $B^k$  as  $B_k$ .

This type of symbolic manipulation is often called the “umbral calculus” and is widely used in formal manipulations in combinatorial problems.

To find more of the Bernoulli numbers we set  $p = 2, 3, \dots$ :

$$p = 2 \quad 0 = C(2, 0)B_0 + C(2, 1)B_1 \rightarrow B_1 = -1/2$$

$$p = 3 \quad 0 = B_0 + 3B_1 + 3B_2 \rightarrow B_2 = 1/6$$

$$p = 4 \quad 0 = B_0 + 4B_1 + 6B_2 + 4B_3 \rightarrow B_3 = 0$$

$$p = 5 \quad 0 = B_0 + 5B_1 + 10B_2 + 10B_3 + 5B_4 \rightarrow B_4 = -1/30$$

Further calculations give

$$B_6 = 1/42 \quad B_8 = -1/30 \quad B_{10} = 5/66$$

A table of  $B_{2k}$  up to  $B_{60}$  can be found in Jordan [25].

We will see in the next chapter that the Bernoulli numbers occur in many places in analysis.

**PROBLEMS**

11.6.1 Compute  $B_6$ .

11.6.2 From the generating function for the reciprocal factorials

$$\sum_{k=0}^{\infty} \frac{t^k}{k!} = e^t$$

derive a relation for the binomial coefficients using

$$e^t \cdot e^t = e^{2t}$$

# 12

## \*INFINITE SERIES

### 12.1 INTRODUCTION

Most books on infinite series discuss convergence, divergence, and “summability” of series at great length, but almost completely neglect the actual computation (summation) of series. One reason for this is the paucity of methods for summing a series in closed form. Of course, if the indefinite summation can be done in closed form and if the series converges, then the infinite series can also be summed by letting the upper limit go to infinity. As an example, we had in Sec. 11.1

$$\sum_{x=1}^n \frac{1}{x(x+1)} = 1 - \frac{1}{n+1}$$

Hence

$$\sum_{x=1}^{\infty} \frac{1}{x(x+1)} = 1$$

In general,

$$\sum_{x=1}^{\infty} \frac{1}{x(x+1) \cdots (x+k-1)} = \frac{1}{k-1} \frac{1}{(k-1)!} \quad k \geq 2 \quad (12.1.1)$$

Frequently a problem in analysis can be reduced to that of evaluating an infinite series. Many times it is more work to evaluate the series than to do the original problem, but sometimes the series representation is an advantage. One of the reasons that the series is preferable is that the error committed, when only a finite number of terms are used, is more easily controlled than the errors made in the direct approach. Thus in the problem of evaluating

$$\int_0^x e^{-t^2} dt$$

for values of  $x$ , say less than 1, we can expand the exponential in an infinite series and integrate term by term to obtain

$$\int_0^x e^{-t^2} dt = \int_0^x \sum_{k=0}^{\infty} \frac{(-1)^k t^{2k}}{k!} dt = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)k!}$$

If we are interested in values of  $x$  less than 1 and want eight figures correct, then taking 11 terms will suffice. This follows from these facts: the series is alternating in sign, the terms are monotone, the series is convergent, and the first neglected term relative to  $x$  has a denominator about  $9.2 \times 10^8$ . Were we to try to estimate the value of the integral by some approximate method of integration taken up later in the book, then the problem of estimating the error would be more difficult.

Occasionally infinite products occur. For example, consider the following problem: around a circle of radius 1 draw an equilateral triangle, around the triangle draw a circle, around the circle draw a square, around the square draw a circle, around the circle draw a regular pentagon, etc. See Fig. 12.1.1. What will happen to the radius of the circle as the process goes to infinity? For the

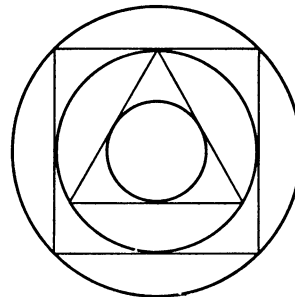


FIGURE 12.1.1



stage of drawing a circle around the  $N$ -sided polygon, let that radius be  $R_N$ . From the picture of one stage we easily see that

$$\begin{aligned}\frac{R_{N-1}}{R_N} &= \cos \frac{180^\circ}{N} \\ R_N &= \frac{R_{N-1}}{\cos(180^\circ/N)} = \frac{R_2}{\prod_{n=3}^N \cos(180^\circ/n)} \\ &= \frac{1}{\prod_{n=3}^N \cos(180^\circ/n)}\end{aligned}$$

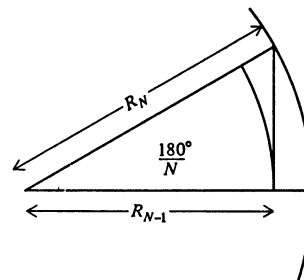


FIGURE 12.1.2

From Table 12.1 we easily see that the sequence of partial products converges to a value not far from 8.7. . . . In a sense we have experimentally determined the convergence and limiting value of the radius of the circle as the number of sides approaches infinity.

Table 12.1

| $N$ | $R$    | $N$ | $R$    | $N$ | $R$    | $N$     | $R$    |
|-----|--------|-----|--------|-----|--------|---------|--------|
|     |        |     |        |     |        | 1,000   | 8.6572 |
|     |        | 10  | 5.2467 | 100 | 8.2831 | 10,000  | 8.6959 |
|     |        | 20  | 6.8367 | 200 | 8.4885 | 20,000  | 8.6981 |
| 3   | 2.0000 | 30  | 7.3997 | 300 | 8.5583 | 30,000  | 8.6993 |
| 4   | 2.8284 | 40  | 7.7017 | 400 | 8.5935 | 50,000  | 8.6996 |
| 5   | 3.4961 | 50  | 7.8900 | 500 | 8.6146 | 100,000 | 8.6996 |
| 6   | 4.0370 | 60  | 8.0184 | 600 | 8.6288 | 200,000 | 8.6996 |
| 7   | 4.4807 | 70  | 8.1118 | 700 | 8.6389 |         |        |
| 8   | 4.8499 | 80  | 8.1826 | 800 | 8.6465 |         |        |
| 9   | 5.2467 | 90  | 8.2383 | 900 | 8.6524 |         |        |

## PROBLEMS

12.1.1 Prove Eq. (12.1.1).

12.1.2 Write the series for

$$Si(x) = \int_0^x \frac{\sin t}{t} dt$$

How many terms do you need in order to compute  $Si(2\pi)$  to eight decimal places?12.2 KUMMER'S<sup>1</sup> COMPARISON METHOD

A method we shall repeatedly use is the following. Given a problem which we cannot solve easily, we look for a “nearby” problem which we can solve, and then we compute or estimate the difference between the two problems. The closer the nearby problem is to the given one, the smaller the difference we compute; hence with correspondingly less effort we can find an accurate answer to the given problem.

If a given series converges rapidly, then there is no serious problem in planning the computation of its sum. If the series converges slowly, then we look for a series with a known sum which converges at “about the same rate” as the given series. The difference between them will therefore converge more rapidly than the original series. Thus we mean by the words *about the same rate* that the difference approaches zero rapidly; and the more rapidly it reaches zero, the more it is at about the same rate. This is known as Kummer's method.

To fix notation, let the given series be

$$S = \sum_{x=r}^{\infty} a_x$$

and suppose we know the sum

$$S' = \sum_{x=r}^{\infty} C_x a_x$$

where  $C_x \rightarrow C$  as  $x \rightarrow \infty$ . Then

$$S = \frac{S'}{C} + \sum_{x=r}^{\infty} \left(1 - \frac{C_x}{C}\right) a_x$$

The term in the parenthesis approaches 0 as  $x \rightarrow \infty$ , and so the series to be computed will converge more rapidly than the original series.

<sup>1</sup> Ernst Eduard Kummer (1810–1893).

For example, consider the series

$$S = \sum_{x=1}^{\infty} \frac{x}{(x^2+1)^2}$$

which converges like  $1/x^3$ . This suggests using

$$S' = \sum_{x=2}^{\infty} \frac{1}{(x-1)x(x+1)} = \frac{1}{4}$$

as a comparison series. Note that we had to start at  $x = 2$  and omit the first term. We write

$$\begin{aligned} S &= S' + (S - S') \\ &= S' + \frac{1}{4} + \sum_{x=2}^{\infty} \left[ \frac{x}{(x^2+1)^2} - \frac{1}{(x-1)x(x+1)} \right] \\ &= \frac{1}{4} + \frac{1}{4} + \sum_{x=2}^{\infty} \frac{x^4 - x^2 - (x^4 + 2x^2 + 1)}{x(x^2-1)(x^2+1)^2} \\ &= \frac{1}{2} - \sum_{x=2}^{\infty} \frac{3x^2 + 1}{x(x^2-1)(x^2+1)^2} = 0.39711677 \dots \end{aligned}$$

The new series converges like  $1/x^5$  (and we could use a similar series for the  $3x^2$  part of the numerator if we wished).

## PROBLEMS

12.2.1 Use  $\sum_1^{\infty} \frac{1}{x(x+1)}$  to approximate  $\sum_1^{\infty} 1/x^2$ .

12.2.2 Approximate result of Prob. 12.2.1 by  $\sum_1^{\infty} \frac{1}{x(x+1)(x+2)}$ . Continue  $k$  steps.

12.2.3 Approximate  $\sum_1^{\infty} 1/x^3$  by  $\sum_2^{\infty} \frac{1}{(x-1)x(x+1)}$

12.2.4 Approximate  $\sum_{x=0}^{\infty} \frac{1}{1+x^2}$  by  $\sum_{x=1}^{\infty} \frac{1}{x(x+1)}$

## 12.3 SOME STANDARD SERIES

Evidently Kummer's comparison method requires an arsenal of known series. There are numerous tables of series such as the classic table of Jolley [24] and the modern Russian translation of Gradshteyn and Ryshik [18].

Another source is the series one learns in the normal course of mathematics, such as the trigonometric and the hyperbolic series. Many of these involve the

Bernoulli numbers introduced in Sec. 11.6. There we defined Bernoulli numbers by the generating function

$$\frac{x}{e^x - 1} = \sum_{k=0}^{\infty} \frac{B_k x^k}{k!}$$

and noted that all  $B_k$  of odd order were 0 except  $B_1 = -1/2$ . Thus

$$\frac{x}{e^x - 1} - \frac{x}{2} = \sum_{k=0}^{\infty} \frac{B_{2k} x^{2k}}{(2k)!}$$

But

$$\begin{aligned} \frac{x}{e^x - 1} + \frac{x}{2} &= \frac{x(e^x + 1)}{2(e^x - 1)} = \frac{x}{2} \left( \frac{e^{x/2} - e^{-x/2}}{e^{x/2} + e^{-x/2}} \right) \\ &= \frac{x}{2} \coth \frac{x}{2} = \sum_{k=0}^{\infty} \frac{B_{2k} x^{2k}}{(2k)!} \end{aligned}$$

or writing  $2x$  in place of  $x$ ,

$$\coth x = 2 \sum_{k=0}^{\infty} B_{2k} \frac{(2x)^{2k-1}}{(2k)!} = \frac{1}{x} + \frac{x}{3} - \frac{x^3}{45} + \frac{2x^5}{945} + \cdots$$

If we put  $x = iz$  in this, we get

$$\begin{aligned} \cot z &= \frac{1}{z} - \frac{z}{3} - \frac{z^3}{45} - \frac{2z^5}{945} + \cdots \\ &= 2 \sum_{k=0}^{\infty} (-1)^k \left[ \frac{(2z)^{2k-1} B_{2k}}{(2k)!} \right] \end{aligned}$$

Now using the trigonometric identity

$$2 \cot 2t = \cot t - \tan t$$

in the form

$$\tan t = \cot t - 2 \cot 2t$$

we get

$$\tan t = \sum_{k=0}^{\infty} \frac{(-1)^{k-1} 2^{2k} (2^{2k} - 1) B_{2k}}{(2k)!} t^{2k-1}$$

Integrating this, we get

$$\ln \cos t = \sum_{k=0}^{\infty} \frac{(-1)^k 2^{2k} (2^{2k} - 1) B_{2k}}{2k(2k)!} t^{2k}$$

Again reverting to a trigonometric identity

$$\frac{1}{\sin t} = \cot t + \tan \frac{t}{2}$$

we get

$$\begin{aligned}\frac{t}{\sin t} &= t \csc t = \sum_{k=0}^{\infty} \frac{(-1)^{k-1}(2^{2k}-2)B_{2k}}{(2k)!} t^{2k} \\ &= 1 + \frac{t^2}{6} + \frac{7}{360} t^4 + \frac{31}{15,120} t^6 + \cdots\end{aligned}$$

Evidently only ingenuity limits how far we can go in this matter. A similar set of numbers, the Euler numbers, gives rise to a number of similar series, where

$$\sec t = \sum_{k=0}^{\infty} E_k \frac{t^k}{k!}$$

## 12.4 THE RIEMANN ZETA FUNCTION

The Riemann<sup>1</sup> zeta function is defined by

$$\zeta(z) = \sum_{k=1}^{\infty} \frac{1}{k^z}$$

and a short table of values (Table 10.6.2) was given in Sec. 10.6. It is a very convenient series for comparison purposes. The values of the zeta function for  $z$  an even integer are known in closed form

$$\zeta(2) = \frac{\pi^2}{6}$$

$$\zeta(4) = \frac{\pi^4}{90}$$

$$\zeta(6) = \frac{\pi^6}{945}$$

$$\zeta(8) = \frac{\pi^8}{9,450}$$

$$\zeta(2n) = \frac{(-1)^{n-1} B_{2n} (2\pi)^{2n}}{2(2n)!}$$

where  $B_{2n}$  are the Bernoulli numbers.

Note that from definition,  $\zeta(n) \rightarrow 1$  as  $n \rightarrow \infty$ , and this indicates how  $B_{2n}$  grows as  $n \rightarrow \infty$ .

From the zeta function it is easy to derive the sums of other series. Using the observation that

$$\frac{\zeta(n)}{2^n} = \frac{1}{2^n} + \frac{1}{4^n} + \frac{1}{6^n} + \cdots = \sum_{k=1}^{\infty} \frac{1}{(2k)^n}$$

<sup>1</sup> Bernhard Riemann (1826–1866).

we have

$$\left(1 - \frac{1}{2^n}\right)\zeta(n) = 1 + \frac{1}{3^n} + \frac{1}{5^n} + \cdots = \sum_{k=1}^{\infty} \frac{1}{(2k-1)^n}$$

and

$$\left(1 - \frac{1}{2^{n-1}}\right)\zeta(n) = 1 - \frac{1}{2^n} + \frac{1}{3^n} - \frac{1}{4^n} + \cdots = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k^n}$$

Again, for  $n$  an even number the sums can be expressed in terms of Bernoulli numbers.

### PROBLEMS

12.4.1 For the series  $\sum_{k=1}^{\infty} 1/(2k)^{2n}$

show that the series for  $n = 1, 2, 3$  are

$$\frac{\pi^2}{24} \quad \frac{\pi^4}{1,440} \quad \frac{\pi^6}{64 \cdot 945}$$

12.4.2 For the series  $\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k^{2n}}$

show that the series for  $n = 1, 2, 3$  are

$$\frac{\pi^2}{12} \quad \frac{7\pi^4}{720} \quad \frac{31\pi^6}{30,240}$$

## 12.5 ANOTHER INTEGRAL EQUATION

This problem, resembling the one in Sec. 9.10, arose in practice. Given  $g(T)$  defined by

$$g(T) = \int_0^{\infty} \frac{f(x)(x/T)^2 e^{x/T}}{(e^{x/T} - 1)^2} dx$$

find  $f(x)$ .

Again we begin by trying  $f(x) = x^n$  and set  $x = Ty$

$$\begin{aligned} g(T) &= \int_0^{\infty} \frac{T^n y^n y^2 e^y}{(e^y - 1)^2} T dy \\ &= T^{n+1} \int_0^{\infty} \frac{y^{n+2} e^{-y}}{(1 - e^{-y})^2} dy \\ &= T^{n+1} \int_0^{\infty} y^{n+2} \sum_{k=1}^{\infty} k e^{-ky} dy \end{aligned}$$

Setting  $ky = z$ ,

$$\begin{aligned} g(T) &= T^{n+1} \left( \sum_{k=1}^{\infty} \frac{1}{k^{n+2}} \right) \int_0^{\infty} e^{-z} z^{n+2} dz \\ &= T^{n+1} (n+2)! \zeta(n+2) \end{aligned}$$

Thus if

$$g(T) = T^{n+1}$$

then

$$f(x) = \frac{x^n}{(n+2)! \zeta(n+2)}$$

Now, if we can represent

$$g(T) = \sum_{n=0}^N a_n T^{n+1}$$

then the solution of the integral equation is

$$f(x) = \sum_{n=0}^N \frac{a_n x^n}{(n+2)! \zeta(n+2)}$$

Evidently the “convergence” of this finite series is rapid in comparison to the original series for  $g(T)$ . We postpone until Chaps. 14, 25, and 28 discussion of various ways we might find the  $a_n$  in the representation of  $g(T)$ .

Consider, now, how this resembles Kummer’s method. We did not try to represent closely the given function  $g(T)$  by a single function  $T^{n+1}$  whose solution we knew. Rather we tried a linear combination of functions  $T^{n+1}$ , each of whose solutions we knew, and then took the same linear combination of the known solutions. We forced all the approximation into the closeness of representation of the original function, and then we operated on the approximate function analytically to get the exact answer to the approximate problem.

In Kummer’s method, and throughout much of numerical methods, we try to replace the given function which we cannot handle analytically by ones which we can, and as in the above case, we operate on these functions with the infinite operators of the problem. In Kummer’s method we are concerned with the representation of the terms far out in the series by suitable approximate terms, and we then compute approximately the difference in our approximation, using mainly terms near the beginning of the series. However, it should be clear that instead of using a single comparison series we could use several simultaneously.

This method of simultaneous approximation in terms of several series differs somewhat from the earlier process of successive approximations, and provides an alternate approach, or if you prefer, an extension of the simple Kummer method.

For example, consider again the series

$$S = \sum_{x=1}^{\infty} \frac{x}{(x^2 + 1)^2}$$

We can use

$$S' = \sum_{x=1}^{\infty} \frac{1}{x(x+1)}$$

$$S'' = \sum_{x=1}^{\infty} \frac{1}{x^3}$$

We therefore write

$$S = aS' + bS'' + S'''$$

where  $a$ ,  $b$ , and  $S'''$  are to be determined. Ignoring for the moment the first term  $x = 1$ , we have for the general term

$$\frac{x}{(x^2 + 1)^2} = \frac{a}{x^3 - x} + \frac{b}{x^3} + \frac{F(x)}{(x^2 + 1)^2 x^3 (x^3 - x)}$$

$$x^7 - x^5 = a(x^7 - 2x^5 + x^3) + b(x^7 + x^5 - x^3 - x) + F(x)$$

Equating like powers of  $x$

$$\begin{array}{lll} x^7: & 1 = a + b & a = -2 \\ x^5: & -1 = 2a + b & b = 3 \end{array}$$

Thus we will have to compute

$$\sum_{x=1}^{\infty} \frac{5x^2 + 3}{(x^2 + 1)^2 x^3 (x^2 - 1)} \approx \frac{5}{x^7}$$

## 12.6 EULER'S METHOD

Another method of summing series numerically is Euler's method, which can be viewed in many ways. The author prefers the following approach.

Consider the finite series

$$\sum_{k=0}^{n-1} a_k t^k \quad (12.6.1)$$

We apply summation by parts and recall that

$$\sum_{k=a}^{b-1} u(k) \Delta v(k) = [u(k)v(k)]_a^b - \sum_{k=a}^{b-1} v(k+1) \Delta u(k)$$



We pick  $u(k) = a_k$  and  $\Delta v(k) = t^k$ . Since we may use any additive constant, we can choose

$$v(k) = \sum_{x=0}^{k-1} t^x = \frac{1-t^k}{1-t}$$

Then

$$\begin{aligned} \sum_{k=0}^{n-1} a_k t^k &= \left[ a_k \frac{1-t^k}{1-t} \right]_0^n - \sum_{k=0}^{n-1} \frac{1-t^{k+1}}{1-t} \Delta a_k \\ &= a_n \frac{1-t^n}{1-t} - \frac{1}{1-t} \sum_{k=0}^{n-1} \Delta a_k + \frac{t}{1-t} \sum_{k=0}^{n-1} t^k \Delta a_k \end{aligned} \quad (12.6.2)$$

but

$$\begin{aligned} \sum_{k=0}^{n-1} \Delta a_k &= (a_1 - a_0) + (a_2 - a_1) + \cdots + (a_n - a_{n-1}) \\ &= a_n - a_0 \end{aligned}$$

and so Eq. (12.6.2) becomes

$$\begin{aligned} \sum_{k=0}^{n-1} a_k t^k &= a_n \frac{1-t^n}{1-t} - \frac{a_n - a_0}{1-t} + \frac{t}{1-t} \sum_{k=0}^{n-1} t^k \Delta a_k \\ &= \frac{a_0}{1-t} - \frac{a_n t^n}{1-t} + \frac{t}{1-t} \sum_{k=0}^{n-1} t^k \Delta a_k \end{aligned} \quad (12.6.3)$$

We apply summation by parts to the third term, noting that it is of the same form as the original series, with  $a_k$  replaced by  $\Delta a_k$  in Eq. (12.6.3).

$$\frac{t}{1-t} \sum_{k=0}^{n-1} t^k \Delta a_k = \frac{t}{1-t} \left( \frac{\Delta a_0}{1-t} - \frac{t^n \Delta a_n}{1-t} + \frac{t}{1-t} \sum_{k=0}^{n-1} t^k \Delta^2 a_k \right)$$

Thus, as a result of two applications of summation by parts, we have obtained

$$\begin{aligned} \sum_{k=0}^{n-1} a_k t^k &= \frac{a_0}{1-t} + \frac{t \Delta a_0}{(1-t)^2} + \frac{t^2}{(1-t)^2} \sum_{k=0}^{n-1} t^k \Delta^2 a_k \\ &\quad - \frac{a_n t^n}{1-t} - \frac{1}{1-t} \frac{\Delta a_n t^{n+1}}{1-t} \end{aligned}$$

After  $r$  applications, the expression (12.6.1) has the form

$$\begin{aligned} \sum_{k=0}^{n-1} a_k t^k &= \frac{1}{1-t} \sum_{i=0}^{r-1} \left( \frac{t}{1-t} \right)^i \Delta^i a_0 + \left( \frac{t}{1-t} \right)^r \sum_{k=0}^{n-1} t^k \Delta^r a_k \\ &\quad - \frac{t^n}{1-t} \sum_{i=0}^{r-1} \left( \frac{t}{1-t} \right)^i \Delta^i a_n \end{aligned}$$

Since the original series converges, given an  $\varepsilon > 0$ , there exists an  $n_0$  such that, for  $n > n_0$ ,  $|a_n| < \varepsilon/2^r$ . Hence, by Sec. 10.3  $|\Delta^i a_n| \leq \varepsilon$ . The last term in the above expression therefore goes to zero as  $n \rightarrow \infty$ , and we have

$$\sum_{k=0}^{\infty} a_k t^k = \frac{1}{1-t} \sum_{i=0}^{r-1} \left(\frac{t}{1-t}\right)^i \Delta^i a_0 + \left(\frac{t}{1-t}\right)^r \sum_{k=0}^{\infty} t^k \Delta^r a_k \quad (12.6.4)$$

It can be shown that if the given series (12.6.1) converges, then the second term on the right also approaches zero as  $r \rightarrow \infty$ . We are then left with the series

$$\sum_{k=0}^{\infty} a_k t^k = \frac{1}{1-t} \sum_{i=0}^{\infty} \left(\frac{t}{1-t}\right)^i \Delta^i a_0 \quad (12.6.5)$$

As an example of the above process, given a series  $\sum_{k=0}^{\infty} u_k$ , if  $u_{k+1}/u_k$  approaches  $t$ , we can write

$$\sum_{k=0}^{\infty} u_k = \sum_{k=0}^{\infty} a_k t^k$$

where  $a_{k+1}/a_k$  approaches 1, and apply Euler's method.

The most frequent case of application is when  $t = -1$ . Euler's method gives, from Eq. (12.6.5),

$$\sum_{k=0}^{\infty} (-1)^k a_k = \frac{1}{2} \sum_{i=0}^{\infty} \frac{(-1)^i}{2^i} \Delta^i a_0 \quad (12.6.6)$$

Sometimes Euler's transformation makes the series converge faster, and sometimes it does not. Consider the following examples.

#### EXAMPLE 12.1

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{2^k}$$

We have

$$\begin{aligned} a_n &= \frac{1}{2^n} \\ \Delta a_n &= \frac{1}{2^{n+1}} - \frac{1}{2^n} = \frac{-1}{2^{n+1}} \\ \Delta^i a_0 &= \frac{(-1)^i}{2^i} \end{aligned}$$

and so by Eq. (12.6.6)

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{2^k} = \frac{1}{2} \sum_{i=0}^{\infty} \frac{(-1)^i}{2^i} \frac{(-1)^i}{2^i} = \frac{1}{2} \sum_{i=0}^{\infty} \frac{1}{4^i}$$

which converges more quickly.

////

**EXAMPLE 12.2**

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{3^k}$$

It is easy to see that

$$\Delta^i a_0 = \frac{(-2)^i}{3^i}$$

and so

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{3^k} = \frac{1}{2} \sum_{i=0}^{\infty} \frac{(-1)^i}{2^i} \frac{(-2)^i}{3^i} = \frac{1}{2} \sum_{i=0}^{\infty} \frac{1}{3^i}$$

which converges somewhat more slowly. ////

**EXAMPLE 12.3** Similarly,

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{4^k} = \frac{1}{2} \sum_{i=0}^{\infty} \left(\frac{3}{8}\right)^i$$

which converges more slowly. ////

The “break point” for the application of Euler’s method to such alternating series is seen to be at ratios of  $|a_i|$  between  $1/2$  and  $1/3$ .

In practice, what is usually done is to sum the first few (say 10) terms directly and then apply Euler’s transformation to the rest.<sup>1</sup>

**PROBLEMS**

**12.6.1** Apply Euler’s method to show that  $\log 2 = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} = \sum_{k=1}^{\infty} \frac{1}{k2^k}$

**12.6.2** Show that

$$\arctan 1 = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} = \frac{1}{2} \sum_{k=0}^{\infty} \frac{k!}{1 \cdot 3 \cdot 5 \cdots (2k+1)}$$

**12.6.3** Add the first eight terms of  $\log 2$  and apply Euler’s method to the rest.

*Ans.*  $1 - 1/2 + 1/3 - \cdots - 1/8 = 0.63452381$ ; Euler part gives  $=(1/2)(0.11724674)$

<sup>1</sup> See J. B. Rosser, Transformations to Speed and Convergence of Series, *J. Res. Natl. Bur. Std.*, vol. 46, 1951, and Bromwich [5].

12.7<sup>1</sup> IMPROVING THE CONVERGENCE OF SEQUENCES

Acceleration of the convergence of a series is only one example of a general technique of numerical analysis, the extrapolation of a slowly convergent or even divergent sequence to its "limit." Such sequences may be obtained as partial sums of series, as successive approximations to the zeros of a function by an iterative process, as estimates of an integral or derivative by finite difference methods using different values of the interval, and in a variety of other ways. In all these sequences, we may regard the terms of the sequence  $S_n$  as estimates of the limit  $S$  with an error  $R_n$ :

$$S_n = S + R_n \quad (12.7.1)$$

If the form of dependence of  $R_n$  upon  $n$  is known, it is frequently possible to estimate the value of  $S$  quite accurately from a few values of  $S_n$ .

If, as frequently happens,  $R_n$  may be approximated by an expression of the form

$$R_n = \sum_{i=1}^k a_i (q_i)^n \quad (12.7.2)$$

a family of nonlinear transformations discovered by Shanks<sup>2</sup> is particularly useful. The basic transformation applied to the partial sums  $S_n$  of a series

$$T(S_n) = \frac{S_{n+1}S_{n-1} - S_n^2}{S_{n+1} - 2S_n + S_{n-1}} \quad (12.7.3)$$

is often very useful. If  $T$  were linear, we would have

$$T(CS_n) = CT(S_n)$$

and

$$T(S_n + U_n) = T(S_n) + T(U_n) \quad (12.7.4)$$

Equation (12.7.4) is not true in general. However, we do have the weaker rule

$$T(S_n + C) = T(S_n) + C$$

If we write

$$S_n = \sum_{m=0}^n a_m$$

<sup>1</sup> This section was suggested by H. C. Thacher, Jr.

<sup>2</sup> D. Shanks, Nonlinear Transformations of Divergent and Slowly Convergent Sequences, *J. Math. Phys.*, vol. 34, pp. 1-42.

then  $T(S_n)$  can be written

$$\begin{aligned} T(S_n) &= \frac{(S_n + a_{n+1})(S_n - a_n) - S_n^2}{S_n + a_{n+1} - 2S_n + S_n - a_n} \\ &= S_n + \frac{a_n a_{n+1}}{a_n - a_{n+1}} \end{aligned} \quad (12.7.5)$$

The repeated application of this transformation is most practically carried out by an algorithm due to Wynn.<sup>1</sup> Let  $\varepsilon_0(S_n) = S_n$ ,  $\varepsilon_{-1}(S_n) = 0$ , and for  $r = 1, 2, 3, \dots$ , let

$$\varepsilon_{r+1}(S_n) = \varepsilon_{r-1}(S_n) + \frac{1}{\varepsilon_r(S_{n+1}) - \varepsilon_r(S_n)}$$

Then the  $\varepsilon_{2k}(S_n)$  are equivalent to the results of applying the  $k$ th Shanks transformation to the sequence  $S_n$ .

As an example of the power of a single application, consider the Leibnitz series

$$\pi = 4 - 4/3 + 4/5 - 4/7 + \dots$$

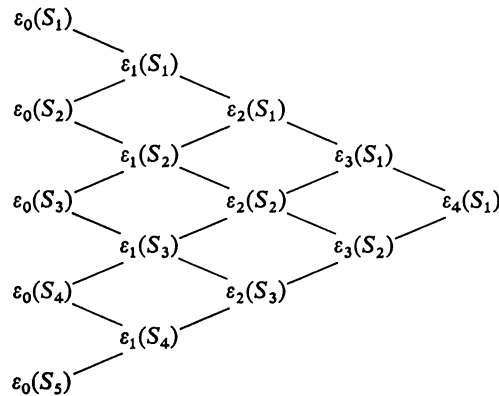
The convergence is so slow that it is practically valueless, but a single application gives the results shown in Table 12.7.

<sup>1</sup> P. Wynn, On a Device for Computing the  $\varepsilon_m(S_n)$  Transformation, *Math. Tables Aids Comp.*, vol. 10, pp. 91–96, 1956.

Table 12.7

| $n$ | $S_n$   | $T(S_n)$ |
|-----|---------|----------|
| 0   | 4.00000 |          |
| 1   | 2.66667 | 3.16667  |
| 2   | 3.46667 | 3.13333  |
| 3   | 2.89524 | 3.14524  |
| 4   | 3.33968 | 3.13968  |
| 5   | 2.97605 | 3.14271  |
| 6   | 3.28374 | 3.14088  |
| 7   | 3.01707 | 3.14207  |
| 8   | 3.25237 | 3.14125  |
| 9   | 3.04184 |          |

The  $\varepsilon_r$  transformations are members of a large family of lozenge, or rhombus, transformations. If the calculations are laid out in the form of a conventional difference table:



the values at the vertices of each rhombus are related by the algorithm. Hence, these transformations are collectively referred to as lozenge algorithms. Another important lozenge algorithm is the  $QD$  algorithm, which may be used for converting power series to equivalent continued fractions. A survey of these algorithms with extensive references was presented by Wynn.<sup>1</sup>

## 12.8 INTEGRALS AS APPROXIMATIONS TO SUMS

The definite integral is defined to be the limit of a sum. It should be evident, therefore, that integrals can be used as approximations to sums. One such actual formula for doing this appears in Sec. 18.9 and may be written as

$$\begin{aligned} f_0 + f_1 + \cdots + f_n &= \int_0^n f(x) dx + 1/2(f_0 + f_n) \\ &\quad - \frac{1}{12}(\Delta f_0 - \Delta f_{n-1}) + \frac{1}{24}(\Delta^2 f_0 + \Delta^2 f_{n-2}) \\ &\quad - \frac{19}{720}(\Delta^3 f_0 - \Delta^3 f_{n-3}) + \frac{3}{160}(\Delta^4 f_0 + \Delta^4 f_{n-4}) + \cdots \end{aligned}$$

<sup>1</sup> P. Wynn, Acceleration Techniques in Numerical Analysis, with Particular Reference to Problems in One Independent Variable, *Proc. IFIPS*, Munich, 1962, pp. 149–156.

A closely related formula is the Euler-Maclaurin formula [63]

$$\begin{aligned} f_0 + f_1 + \cdots + f_n &= \int_0^n f(x) dx + \frac{1}{2}(f_0 + f_n) - \frac{1}{12}(f'_0 - f'_n) \\ &+ \frac{1}{720}(f'''_0 - f'''_n) - \frac{1}{30,240}(f^{(5)}_0 - f^{(5)}_n) + \cdots \\ &+ \frac{B_{2m}}{(2m)!}(f^{(2m-1)}_0 - f^{(2m-1)}_n) + \cdots \end{aligned}$$

where, again, the  $B_{2m}$  are the Bernoulli numbers.

In both formulas when dealing with infinite series, the upper limits are infinite and half the terms drop out.

## 12.9 THE DIGAMMA FUNCTION

For the final topic of this chapter, recall the formula

$$\int^x t^m dt = \frac{x^{m+1}}{m+1} + C$$

which does not apply when  $m = -1$ . Indeed, when  $m = -1$ , the integral can be said to define a new function,  $\log x$ .

Similarly, the summation formula

$$\sum_{t=1}^{x-1} t^{(m)} = \frac{x^{(m+1)}}{m+1} + C$$

does not apply when  $m = -1$  and correspondingly can be said to define a new function, called the digamma function and denoted by  $F(x)$ :

$$F(x) = \sum_{r=1}^x \frac{1}{r} - \gamma$$

where  $\gamma = \text{Euler's constant} = 0.5772156649 \dots$  and hence  $F(0) = -\gamma$ . This formula applies when  $x$  is an integer. Another form,

$$F(x) = \sum_{r=1}^{\infty} \frac{x}{r(r+x)} - \gamma \quad F(0) = -\gamma$$

provides a means of extending the definition of  $F(x)$  to noninteger values. We need to show that this new definition is consistent with the old one. This amounts to showing that  $F(x)$  satisfies the equation

$$\Delta F(x) = \frac{1}{x+1}$$

(This is analogous to the process of showing that a function satisfies an integration formula by showing that the derivative of the function is the integrand.) We have

$$\begin{aligned}\Delta F(x) &= \sum_{r=1}^{\infty} \left[ \frac{x+1}{r(r+x+1)} - \frac{x}{r(r+x)} \right] \\ &= \sum_{r=1}^{\infty} \frac{1}{(r+x)(r+x+1)} \\ &= \sum_{r=1}^{\infty} \left[ \frac{1}{r+x} - \frac{1}{r+x+1} \right] \\ &= \frac{1}{x+1}\end{aligned}$$

The following relation exists between the digamma function and the natural logarithm:

$$F(x) = \lim_{n \rightarrow \infty} \left[ \log(x+n+1) - \frac{1}{x+1} - \frac{1}{x+2} - \cdots - \frac{1}{x+n} \right]$$

Another relationship between known functions is

$$\frac{d}{dx} \log[\Gamma(1+x)] = F(x)$$

If we differentiate again, we get the so-called trigamma function

$$\frac{d^2}{dx^2} \log[\Gamma(1+x)] = \sum_{r=1}^{\infty} \frac{1}{(r+x)^2} = F'(x)$$

the tetragamma function

$$\frac{d^3}{dx^3} \log[\Gamma(1+x)] = -2 \sum_{r=1}^{\infty} \frac{1}{(r+x)^3} = F''(x)$$

the pentagamma function

$$\frac{d^4}{dx^4} \log[\Gamma(1+x)] = 6 \sum_{r=1}^{\infty} \frac{1}{(r+x)^4} = F'''(x)$$

etc.

These functions have been tabulated [47] and can be used to sum series whose general coefficient is a rational function. It should be recognized that much of systematic integration is based on the integration of rational functions and that this introduces the new functions  $\ln x$  and  $\arctan x$ . Similarly the summation of rational terms leads to the corresponding functions.



For example, supposing we have

$$S = \sum_{x=1}^{\infty} \frac{P_{n-2}(x)}{P_n(x)}$$

where  $P_n(x)$  is a polynomial of degree  $n$  and  $P_{n-2}(x)$  is at least two degrees less than  $P_n(x)$ . Using partial fractions, we can write [where the  $a_i$  are the zeros of  $P_n(x)$ ]

$$S = \sum_{x=1}^{\infty} \left[ \frac{A_1}{x-a_1} + \frac{A_2}{x-a_2} + \cdots + \frac{A_k}{x-a_k} + \frac{B_1}{(x-a_1)^2} + \frac{B_1}{(x-a_2)^2} \right. \\ \left. + \cdots + \frac{B_k}{(x-a_k)^2} + \cdots + \frac{M_1}{(x-a_1)^m} + \frac{M_2}{(x-a_2)^m} + \cdots + \frac{M_k}{(x-a_k)^m} \right]$$

We cannot rearrange the first group of terms since individually the series diverge. However, it is easy to see that  $\sum_{i=1}^k A_i = 0$ ; hence we can write

$$\left[ \left( \frac{A_1}{x-a_1} - \frac{A_1}{x} \right) + \left( \frac{A_2}{x-a_2} - \frac{A_2}{x} \right) + \cdots + \left( \frac{A_k}{x-a_k} - \frac{A_k}{x} \right) \right]$$

without changing the value of the term. We now have the series in a form that we can rearrange easily. As an example, consider [47]

$$S = \sum_{x=1}^{\infty} \frac{1}{(4x+2)(4x+1)(4x+3)^2} \\ = \sum_{x=1}^{\infty} \left[ -\frac{1}{4x+2} + \frac{1}{4} \cdot \frac{1}{4x+1} + \frac{3}{4} \cdot \frac{1}{4x+3} + \frac{1}{2} \cdot \frac{1}{(4x+3)^2} \right] \\ = 1/4F(1/2) - 1/16F(1/4) - 3/16F(3/4) + 1/32F'(3/4)$$

# 13

## DIFFERENCE EQUATIONS

### 13.1 INTRODUCTION

In the finite difference calculus the analogs of differential equations are difference equations. It might be supposed that corresponding to the differential equation

$$y' + 2y = x$$

there would be the difference equation

$$\Delta y + 2y = x$$

It is a curious fact that this is *not* the analogous form; instead we use

$$y(x+1) + y(x) = x$$

which follows from writing

$$\Delta y(x) = y(x+1) - y(x)$$

It is easy to pass from one form to the other, and they are equivalent in this sense. How we would go about solving the two forms, however, would be different.

One advantage of the form using different arguments, rather than the  $\Delta$  operator, is that it more clearly reveals the nature of the problem. The equation

$$\Delta^2 y + 2\Delta y + y = f(x)$$

looks as if it were of second order, but writing it in the function-value form

$$y(x+2) - 2y(x+1) + y(x) + 2[y(x+1) - y(x)] + y(x) = f(x)$$

leads to

$$y(x+2) = f(x)$$

which is trivial. It is the maximum of the difference in the arguments of the unknown function that determines the order of the equation.

The close analogy between the difference and differential equations will be exploited to shorten the treatment. We will also look at only the simpler problems in the field.

Among the various ways of approaching a field, one can give the general theory or instead give only a few well-chosen examples from which the reader can easily understand how the general theory goes. There are many occasions when the second approach is more effective, and in this chapter we shall adopt it. Thus the chapter is mainly a sequence of carefully chosen examples which illustrate how to solve typical difference equations of the forms we need.

### 13.2 FIRST-ORDER DIFFERENCE EQUATIONS WITH CONSTANT COEFFICIENTS

To solve the first-order differential equation

$$y' + y = x$$

we first consider the homogeneous equation

$$y' + y = 0$$

To solve this, we *guess* at a solution of the form

$$y = e^{mx}$$

and get

$$e^{mx}(m+1) = 0$$

$$m = -1$$

$$y = e^{-x}$$

and the general solution of the homogeneous equation is

$$y(x) = Ce^{-x}$$

For the complete equation we again *guess*, this time at one particular solution. In this case it is reasonable to suspect that the right-hand side could only arise from a function like

$$y = ax + b$$

Trying this, we get

$$a + ax + b \equiv x$$

which determines  $a = 1$  and  $b = -1$ . Adding this solution to the general solution of the homogeneous equation, we have the general solution of the original equation

$$y = Ce^{-x} + (x - 1)$$

All this should be familiar from the calculus course, and follows from the linearity of the operation of differentiation and the linearity of the equation. We expect, therefore, that the solution of the corresponding difference equation will be similar.

Given the difference equation

$$y(x+1) + y(x) = x$$

we first solve the homogeneous equation

$$y(x+1) + y(x) = 0$$

We guess at

$$y(x) = a^x$$

where we have used the conventional notation of writing  $a = e^m$ . The result of substituting this into the homogeneous equation is

$$a^x(a+1) = 0$$

$$a = -1$$

$$y(x) = C(-1)^x$$

For the particular solution of the complete equation with the right-hand side  $x$ , we again *guess* at the form

$$y(x) = ax + b$$

from which we get

$$a(x+1) + b + ax + b \equiv x$$

$$a = 1/2 \quad b = -1/4$$

and the complete solution is

$$y = C(-1)^x + \frac{2x-1}{4}$$

**PROBLEMS**

13.2.1 Solve the equation  $y_{n+1} - y_n = n + 1$   $y(0) = 0$ .

$$\text{Ans. } y = \frac{n(n+1)}{2}$$

13.2.2 Solve  $y_{n+1} + y_n = n^2$   $y(0) = 0$ .

13.2.3 Explain what you must do to solve  $y_{n+1} - ay_n = a^n$

$$\text{Ans. } y = (n + C)a^{n-1}$$

### 13.3 THE GENERAL FIRST-ORDER LINEAR DIFFERENCE EQUATION

We next consider

$$y(x+1) + A(x)y(x) = B(x)$$

where  $A(x)$  and  $B(x)$  are known functions. The homogeneous equation is

$$y(x+1) + A(x)y(x) = 0$$

thus

$$\begin{aligned} y(x+1) &= -A(x)y(x) = A(x)A(x-1)y(x-1) = \cdots \\ &= (-1)^{x+1}A(x)A(x-1)\cdots A(0)y(0) \end{aligned}$$

where  $y(0)$  is an arbitrary constant, say  $C$ . Thus a solution is

$$y(x+1) = \prod_{k=0}^x [-A(k)]$$

Let this solution be labeled  $u(x)$ .

As in the case of differential equations, we set

$$y(x) = u(x)v(x)$$

where  $u(x)$  is the above solution. Putting this into the complete equation, we get

$$u(x+1)v(x+1) + A(x)u(x)v(x) = B(x)$$

Substituting  $v(x+1) = v(x) + \Delta v(x)$ , we get

$$v(x)[u(x+1) + A(x)u(x)] + u(x+1)\Delta v(x) = B(x)$$

The term within the square bracket is zero, since  $u(x)$  is a solution of the homogeneous equation. Thus

$$\Delta v(x) = \frac{B(x)}{u(x+1)}$$

Summing this over the variable  $x$  from  $x = 0$  to  $x = x$ , we get

$$v(x+1) = \sum_{k=0}^x \frac{B(k)}{u(k+1)} + C$$

where  $C$  is an arbitrary constant.

This is exactly analogous to the case of differential equations and is known as “the variation of parameters method.”

### PROBLEMS

Solve the following problems.

13.3.1  $ny(n+1) - (n+1)y(n) = 1$

13.3.2  $y(n+1) - n^2y(n) = 0, y(1) = 1$

*Ans.*  $y(n+1) = (n!)^2$

13.3.3  $(n+1)y(n+1) - ny(n) = n^2$

*Ans.*  $y(n+1) = \frac{C}{n+1} + \frac{2n^2 - 3n + 1}{6}$

13.3.4  $y(n+1) - 3ny(n) = 3^n$

13.3.5  $y(n+1) - ny(n) = n \quad (n > 0)$

13.3.6  $y(n+1) - (n+1)y(n) = 2^n(n-1)$

### 13.4 THE FIBONACCI EQUATION

Leonardo of Pisa (1180–1250) was interested in the famous second-order difference equation with constant coefficients

$$y_{n+1} = y_n + y_{n-1} \quad y_0 = 0 \quad y_1 = 1$$

where we have now adopted the subscript notation.

We again try

$$y_n = a^n$$

and get

$$a^{n-1}(a^2 - a - 1) = 0$$

$$a^2 - a - 1 = 0$$

$$a = \frac{1 \pm \sqrt{5}}{2}$$

Hence the general solution is

$$y_n = C_1 \left( \frac{1 + \sqrt{5}}{2} \right)^n + C_2 \left( \frac{1 - \sqrt{5}}{2} \right)^n$$

To satisfy the two initial conditions, we have available the two constants  $C_1$  and  $C_2$ . We get the two following equations for them:

$$\begin{aligned}y_0 &= C_1 + C_2 = 0 \\y_1 &= C_1 \left( \frac{1 + \sqrt{5}}{2} \right) + C_2 \left( \frac{1 - \sqrt{5}}{2} \right) = 1\end{aligned}$$

This gives

$$C_1 = \frac{1}{\sqrt{5}} = -C_2$$

and the solution is

$$y_n = \frac{1}{\sqrt{5}} \left( \frac{1 + \sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left( \frac{1 - \sqrt{5}}{2} \right)^n$$

Note that the square root cancels out (as it must since from the original equation all the Fibonacci numbers are integers).

Mathematically speaking this is the solution, but practically it is generally easier to compute the numbers from the equation than from the solution. Thus we easily get

$$\begin{aligned}y_2 &= y_1 + y_0 = 1 + 0 = 1 \\y_3 &= y_2 + y_1 = 1 + 1 = 2 \\y_4 &= y_3 + y_2 = 2 + 1 = 3 \\y_5 &= y_4 + y_3 = 3 + 2 = 5 \\y_6 &= y_5 + y_4 = 5 + 3 = 8 \\&\dots\dots\dots\end{aligned}$$

An alternate approach to the Fibonacci numbers uses generating functions. If we multiply the defining equation by  $t^n$  and sum  $n = 1, 2, \dots$ , we get

$$\frac{1}{t} \left( \sum_{n=1}^{\infty} t^{n+1} y_{n+1} + t y_1 \right) - y_1 = \sum_{n=1}^{\infty} t^n y_n + t \sum_{n=1}^{\infty} t^{n-1} y_{n-1}$$

Setting

$$\sum_{n=1}^{\infty} t^n y_n = Y(t)$$

we have the following equation for the generating function  $Y(t)$

$$\begin{aligned}\frac{1}{t} Y(t) - y_1 &= Y(t) + tY(t) \\ Y(t) \left( \frac{1}{t} - 1 - t \right) &= y_1 = 1 \\ Y(t) &= \frac{1}{1/t - 1 - t} = \frac{t}{1 - t - t^2}\end{aligned}$$

If this rational function is broken up into two linear factors and these are suitably divided out, then the earlier expression for  $y_n$  is obtained.

#### PROBLEMS

13.4.1 Divide out the generating function to find the first five Fibonacci numbers.

13.4.2 Find a formula for the sum of the first  $N$  Fibonacci numbers.

$$\text{Ans. } \sum_1^N y_n = y_{N+1} + y_N - 1$$

13.4.3 Carry out the steps in the last paragraph in this section.

### 13.5 ANOTHER EXAMPLE OF A SECOND-ORDER LINEAR EQUATION

As a second example, consider the equation

$$x_{n+1} - 2x_n \cos \phi + x_{n-1} = 0$$

where  $\phi$  is some constant. We try

$$\begin{aligned}x_n &= a^n \\ a^2 - 2a \cos \phi + 1 &= 0 \\ a &= \cos \phi \pm \sqrt{\cos^2 \phi - 1} \\ &= \cos \phi \pm i \sin \phi \\ &= e^{i\phi}, e^{-i\phi}\end{aligned}$$

Hence the solution is

$$x_n = C_1 e^{in\phi} + C_2 e^{-in\phi}$$

or, if you prefer,

$$x_n = \bar{C}_1 \cos n\phi + \bar{C}_2 \sin n\phi$$

This can, of course, be checked by direct substitution into the difference equation.



## PROBLEMS

Solve

$$13.5.1 \quad y_{n+1} - 2y_n \cosh \phi + y_{n-1} = 0$$

$$13.5.2 \quad y_{n+1} + 2ay_n - b^2 y_{n-1} = 0$$

$$13.5.3 \quad y_{n+1} = a_1 y_n + a_2 y_{n-1} \quad y_0 = 0 \quad y_1 = a$$

## 13.6 AN EXAMPLE OF A SYSTEM OF EQUATIONS

A system of two first-order difference equations is equivalent to a single second-order equation. Often all that is needed is the behavior of the solution, and we now give an example of how to find the behavior without actually solving the problem. Consider the system of equations

$$\begin{aligned} p_{n+1} &= p_n + Kq_n \\ q_{n+1} &= p_n + q_n \end{aligned} \quad (13.6.1)$$

We know that the solutions will be of the form

$$\begin{aligned} p_n &= Aa_1^n + Ba_2^n \\ q_n &= Ca_1^n + Da_2^n \end{aligned} \quad (13.6.2)$$

with some constraints on the four coefficients  $A, B, C$ , and  $D$ . We assume  $|a_1| > |a_2|$ . We can write

$$\frac{p_{n+1}}{q_{n+1}} = \frac{p_n + Kq_n}{p_n + q_n} = \frac{p_n/q_n + K}{p_n/q_n + 1} \quad (13.6.3)$$

But

$$\frac{p_n}{q_n} = \frac{Aa_1^n + Ba_2^n}{Ca_1^n + Da_2^n} = \frac{A + B(a_2/a_1)^n}{C + D(a_2/a_1)^n} \rightarrow \frac{A}{C}$$

as  $n \rightarrow \infty$ . Let  $A/C = r$ . Then in the limit equation, Eq. (13.6.3) gives

$$r = \frac{r + K}{r + 1}$$

This is the same as

$$r^2 + r = r + K$$

$$r = \sqrt{K} = \lim_{n \rightarrow \infty} \frac{p_n}{q_n}$$

Thus we have a way of computing square roots regardless of the initial choices  $p_0$  and  $q_0$  (not both zero).

## PROBLEMS

13.6.1 In the above example, discuss the cases where

$$\begin{aligned}a_1 &= a_2 \\ a_1 &= -a_2\end{aligned}$$

13.6.2 Discuss the case

$$\begin{aligned}p_{n+1} &= ap_n + bq_n \\ q_{n+1} &= cp_n + dq_n\end{aligned}$$

13.6.3 Discuss the possibilities of the starting values for  $p_0$  and  $q_0$  eliminating  $a_1$ , that is,  $A = C = 0$ , with due regard to roundoff.

## 13.7 A SYSTEM OF EQUATIONS WITH VARIABLE COEFFICIENTS

Consider finding the values of the two integrals

$$J(n) = \int_0^\infty x^n e^{-x} \sin x \, dx$$

$$n \geq 0$$

$$K(n) = \int_0^\infty x^n e^{-x} \cos x \, dx$$

We shall be particularly interested in showing that  $J(4k + 3) = 0$ , because using the transformation  $x = t^{1/4}$ , we get the integral

$$4J(4k + 3) = \int_0^\infty t^k (e^{-t^{1/4}} \sin t^{1/4}) \, dt = 0 \quad k = 0, 1, \dots$$

This shows that the kernel in the parentheses has *all* its moments equal to zero, but is not identically zero. Thus the moments of a function do not uniquely determine the function.

The second purpose of this example is to show how a system of equations with variable coefficients can sometimes be reduced to a system with constant coefficients by a suitable choice of notation.

Using integration by parts on both integrals, we get the pair of simultaneous linear difference equations

$$J(n) = \frac{n}{2} [J(n-1) + K(n-1)]$$

$$K(n) = \frac{n}{2} [-J(n-1) + K(n-1)]$$

Standard integration tables give  $J(0) = K(0) = 1/2$ .

The variable coefficient  $n/2$  suggests that  $J(n)$  has a factor  $n/2$ ,  $J(n-1)$  has a factor  $(n-1)/2$ , etc., which suggests that  $J(n)$  behaves like  $n!/2^n$ .

Thus we are led to the transformation

$$J(n) = \frac{n!}{2^n} j(n)$$

$$K(n) = \frac{n!}{2^n} k(n)$$

which leads to the system

$$j(n) = j(n-1) + k(n-1) \quad j(0) = 1/2$$

$$k(n) = -j(n-1) + k(n-1) \quad k(0) = 1/2$$

We reduce this to a single second-order equation by writing the top equation as

$$j(n+1) = j(n) + k(n)$$

and using the second equation of the pair, we eliminate  $k(n)$

$$j(n+1) = j(n) + -j(n-1) + k(n-1)$$

Now using the first equation, we eliminate  $k(n-1)$

$$j(n+1) - 2j(n) + 2j(n-1) = 0 \quad j(0) = 1/2 \quad j(1) = 1$$

The general solution is (where  $i = \sqrt{-1}$ )

$$j(n) = C_1(1+i)^n + C_2(1-i)^n$$

The initial conditions give

$$C_1 + C_2 = 1/2$$

$$(1+i)C_1 + (1-i)C_2 = 1$$

which leads to

$$C_1 = \frac{1-i}{4} \quad C_2 = \frac{1+i}{4}$$

and finally

$$j(n) = \frac{1}{2} [(1+i)^{n-1} + (1-i)^{n-1}]$$

or

$$J(n) = \frac{n!}{2^{n+1}} [(1+i)^{n-1} + (1-i)^{n-1}]$$

(which is actually a real number).

To find  $J(4k+3)$ , we observe that

$$(1+i)^4 = 1 + 4i + 6i^2 + 4i^3 + i^4 = -4 = (1-i)^4$$

hence

$$J(4k+3) = \frac{(4k+3)!}{2^{4k+4}} \{(-4)^k [(1+i)^2 + (1-i)^2]\} = 0$$

### PROBLEMS

13.7.1 Show that  $K(4k+1) = 0$

13.7.2 If

$$I_k(\phi) = \int_0^\pi \frac{\cos k\theta - \cos k\phi}{\cos \theta - \cos \phi} d\theta$$

show that for  $k$  an integer,  $I_k(\phi)$  satisfies

$$I_{k+2}(\phi) - 2 \cos \phi I_{k+1}(\phi) + I_k(\phi) = 0 \quad I_1(\phi) = \pi$$

and hence

$$I_k(\phi) = \frac{\pi \sin k\phi}{\sin \phi}$$

13.7.3 Solve the system of equations in this section as a system.

## 13.8 SECOND-ORDER RECURRENCE RELATIONS

Many of the special functions that arise in practice satisfy second-order linear-recurrence relations. For example,

$$\sin n\theta: \quad \sin(n+1)\theta - 2 \cos \theta \sin n\theta + \sin(n-1)\theta = 0$$

$$\cos n\theta: \quad \cos(n+1)\theta - 2 \cos \theta \cos n\theta + \cos(n-1)\theta = 0$$

$$T_n(x): \quad T_{n+1}(x) - 2xT_n(x) + T_{n-1}(x) = 0$$

where  $T_n(x)$  is the Chebyshev polynomial of order  $n$  (see Chaps. 28 and 29). Not all special functions satisfy such equations with constant coefficients. Most of the special functions have equations with variable coefficients; for example,

$$f_{n+1}(z) - \frac{2n}{z} f_n(z) + f_{n-1}(z) = 0$$

is the equation for the Bessel functions  $J_n(z)$  and  $Y_n(z)$  of the first and second kinds. The variable is, of course,  $n$ , and not  $z$ .

It is reasonable to expect that locally the solutions of an equation with variable coefficients will behave much like the solutions of an equation with constant coefficients, provided the constants are some reasonable local averages of the variable coefficients. Assuming that this is so, we recall that the solution was typically the sum of two solutions and of the form

$$y_n = C_1 a_1^n + C_2 a_2^n$$

Supposing that  $|a_1| > |a_2|$ , then even if we started with  $C_1 = 0$ , roundoff would bring in some of the first solution, and in time this would dominate so that the second solution would not be visible

$$y_n = C_1 a_1^n \left[ 1 + \frac{C_2}{C_1} \left( \frac{a_2}{a_1} \right)^n \right] \rightarrow C_1 a_1^n$$

Thus because of roundoff noise the calculation of the second solution is not practical this way.

This is what usually happens in practice—we want the solution that is growing less rapidly than the other. For example, the Bessel function  $J_n(x)$  decreases as a function of  $n$  in the long run, while  $Y_n(x)$  grows toward infinity. It is obvious that the way to counteract this effect is to compute from high values of  $n$  to low values so that the desired solution will grow and the unwanted one will decay. Thus starting out with a sufficiently high index  $n$ , we set the  $(n+1)$ st solution to zero and the  $n$ th one to some value, say 1. Recurring down, we see the numbers grow (there may be local oscillation, but the amplitude will grow) until we get down to the first value. We will not have  $J_0(x)$  but rather some constant multiple of it, and our problem is to find this constant. In the case of the Bessel function it is customary to use the known identity

$$J_0(z) + 2 \sum_{m=1}^{\infty} J_{2m}(z) = 1$$

to determine this constant. Thus the recurrence has given us not only  $J_0(x)$  but also many of the values  $J_n(x)$  (not all the way up to the starting value but quite a way up *provided* we started high enough in  $n$ ).

This is an important fact to realize; it means that when we are presented with an expansion of a function in terms of a special function, we may many times find all the function values (for a given  $x$ ) with a comparatively small amount of computation. Thus much of classical mathematics becomes practical when this device is used.

We have not discussed the way to pick the starting index  $n$ , as this is a complicated topic of some specialization, nor have we made the treatment rigorous.

When it is necessary to do some such recurrence, the reader can consult the literature.<sup>1</sup>

In the case of the trigonometric difference equations it has been observed that it is better to use the pair of first-order equations than the second-order equation. Furthermore, experience shows that the use of  $\cos \theta = 1 - \phi(\theta)$  is preferable. Thus the equations to be used are

$$\sin(x + \Delta x) = \sin x - \phi(\Delta x)\sin x + \cos x \sin \Delta x$$

$$\cos(x + \Delta x) = \cos x - \phi(\Delta x)\cos x - \sin x \sin \Delta x$$

These are very useful in practice, especially when converting point by point from polar to rectangular coordinates. Experience shows that the accuracy is better than  $\pm 3$  in the sixth significant figure in an eight-digit computation *provided* the equations are reinitialized every 100 steps. If  $\pm 3$  in the fifth is acceptable, then reinitialize every 500 steps.

<sup>1</sup> See the classic papers by F. W. J. Olver, Numerical Solution of Second-Order Linear Difference Equations, *J. Res., Natl. Bur. Std.*, vol. 71, nos. 2 and 3, pp. 111–129, Apr.–Sept. 1967; and Bounds for the Solutions of Second-Order Linear Difference Equations, *J. Res., Natl. Bur. Std.*, vol. 71, no. 4, Oct.–Dec. 1967.



PART II

# Polynomial Approximation-Classical Theory





## POLYNOMIAL INTERPOLATION

## 14.1 ORIENTATION

Speaking in theatrical terms, we have just finished Act I in which various ideas and techniques are introduced and used to meet a number of short-range goals. In the second act, Part II of this book, the individual ideas and techniques will reappear in various ways, and a number of new ideas will also make their appearance. It will also include the further development of the basic themes with more of the classic “unity, emphasis, and coherence.” Act III, in its turn, will give some new views of what happens in Act II, as well as introduce a few new themes. Regretfully for the analogy (and for the book), Parts IV and V are not the dramatic conclusion in which all the loose ends are gathered together, but rather they drift into a collection of miscellaneous topics needed to round out the whole. Thus they lack much of the central unity of Parts II and III, and only a few of the earlier themes are picked up and developed further.

In Part I we were almost exclusively concerned with finite, discrete problems and faced the infinite only in the very mild form of an infinite series. Thus the finite nature of the machine occurred mainly in the form of roundoff. In Part II we begin with interpolation, the simplest of the problems in which truncation

error occurs, and we proceed to more difficult problems such as integration. We will also introduce (Chaps. 25 to 30) a pair of new ideas as to what is regarded as a solution to the problem of approximating an infinite operation with a finite amount of arithmetic on a finite computing machine.

We are more concerned with methods than with results, and therefore the results are often scattered in various chapters. For example, "Interpolation" is the title of this chapter, but it is also treated in Chaps. 15 and 17 through 20; the Index provides the cross references to subject material, while the chapters tend to be organized around the methods and ideas used.

## 14.2 INTERPOLATION

The first problem in interpolation that a person usually meets is that of "reading between the lines" of a published table, say a trigonometric or logarithmic table. In this process it is customary to use a straight line that passes through two adjacent table points—exactly the same as we did in the false-position method (Sec. 4.4). If we use the straight line to compute the values of the function in the interval  $a \leq x \leq b$ , then it is called "interpolation," while if the value to be computed from the line lies outside the interval, then it is called "extrapolation." The formulas in both cases are the same

$$\begin{aligned} f(x) &= f(a) + \frac{x-a}{b-a} [f(b) - f(a)] \\ &= \frac{(b-x)f(a) + (x-a)f(b)}{b-a} \end{aligned}$$

it is only the value of  $x$  that differs.

In the case of Newton's method for finding the real zeros of a function (Sec. 4.6), we used *both* the value of the function and the value of the first derivative at the point to determine the approximating line and its zero. The formula was

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

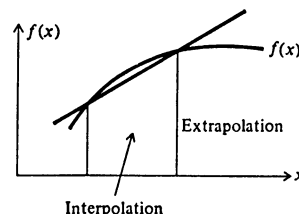


FIGURE 14.2.1

To generalize, we can use as our samples *both* function values and values of various orders of derivatives to determine a suitable-degree polynomial that will be used as the basis for interpolation. We have called this process “analytic substitution” (Sec. 4.6)—for the function we cannot handle we substitute an analytic approximation that we can.

The samples we use can arise directly from measurement, or they can come from computing values of a function that we cannot handle and are forced to sample. In the first case we generally have little or no control over the samples we are given (the information), and the samples are generally equally spaced in the independent variable. In the second case we can select what we will use, and this freedom is applied in later chapters, especially in Chap. 19 where we will use sampling places that in practice would be awkward to measure but are easy to compute from a function (on a computer, but not by hand).

Because of the wide variety of possible combinations of sampling places, and of values of functions and derivatives that are possible (and actually do occur in practice), it is hopeless to try to give a formula for each specific situation. Instead we give a general approach which finds the particular formula that takes advantage of whatever information is available. We also need a broad approach within which we can compare various formulas, so that we can both pick out the right one to use and compare alternate approaches to a situation.

The plan is, therefore, to examine the use of information in these forms:

- Function values only—Chaps. 14 and 15
- Values of the derivatives—Chap. 17
- Differences of function values—Chap. 18
- Arbitrarily placed samples—Chap. 19

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Once we have decided on the information we are going to use (position as well as function and derivative values), then we face the problem of picking the class of approximation functions to be used. In Part II we will use polynomials exclusively, and this is the classical approach. In Part III we will use sines and cosines, which is the more modern approach. Exponential approximation occurs in Part IV.

Next we need to consider the criteria we will use to pick out the particular member of the class to use. We have so far used polynomials that exactly matched (agreed) with the information in the samples (within roundoff). In Chaps. 25 to 30, we will use two other criteria, least squares and the Chebyshev, or minimax, criteria.

Lastly, where shall we apply the test for matching? This question we will postpone for some time, though we faced it in finding zeros when we asked whether it was the closeness of the function to zero or the closeness of the com-